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1 Preliminary: Interaction of Ions with Matter

Charged particles traversing matter lose energy primarily through electromagnetic interactions with atomic electrons and nuclei. The fundamental concepts include the *stopping power*, *range*, and *dose deposition*.

1.1 Types of Interactions

Based on the impact parameter b relative to the atomic radius a (classical electron radius context), interactions are classified as follows:

1.1.1 Soft Collisions ($b \gg a$)

The passing ion's Coulomb field perturbs the atom as a whole:

- Excitation to higher electronic levels.
- Ionization by ejecting weakly bound valence electrons.
- Energy transfer is small ($\sim \text{eV}$).

1.1.2 Hard Collisions ($b \sim a$)

Interaction with an individual atomic electron (approximated as free). Substantial kinetic energy is transferred:

- Ejected electrons (secondary or δ -rays) deposit energy along their tracks.
- Possible emission of characteristic X-rays or Auger electrons.

The interaction probability varies by particle type.

1.1.3 Coulomb Interaction with Nuclei ($b \ll a$)

Primarily for light particles like electrons:

- In $\approx 97\%$ of cases, nearly elastic scattering with negligible energy loss.
- In $2 - 3\%$ of cases, large energy loss via bremsstrahlung emission (inelastic radiative interaction).

Bremsstrahlung probability scales with Z^2 (atomic number of medium) and kinetic energy (important for MeV energies and high- Z materials). For antimatter (e.g., positrons), in-flight annihilation produces two photons.

1.1.4 Nuclear Interactions (Heavy Ions)

For heavy charged particles with $T \gtrsim 100 \text{ MeV}$:

- Inelastic collisions with nuclei lead to intranuclear cascades, ejecting nucleons.
- Excited nuclei de-excite via evaporation of low-energy nucleons and γ -rays.
- Dose contribution is small ($< 1 - 2\%$), and energy may not be deposited locally.

1.2 Stopping Power

Stopping power is the mean rate of energy loss per unit path length:

$$S(T) = -\frac{dT}{dx}, \quad (1)$$

where T is kinetic energy. Units: MeV/cm or J/m.

The mass stopping power normalizes by density ρ :

$$\frac{S(T)}{\rho} = -\frac{1}{\rho} \frac{dT}{dx}, \quad (2)$$

with units $\text{MeV}\cdot\text{cm}^2/\text{g}$ or $\text{J}\cdot\text{m}^2/\text{kg}$.

1.2.1 Collision vs. Radiative Components

The total mass stopping power splits into:

$$\left(\frac{S(T)}{\rho}\right) = \left(\frac{S(T)}{\rho}\right)_c + \left(\frac{S(T)}{\rho}\right)_r, \quad (3)$$

where subscript c denotes collision (local dose deposition) and r denotes radiative (energy carried away by photons, e.g., bremsstrahlung).

The collision part further divides into soft and hard contributions:

$$\left(\frac{S(T)}{\rho}\right)_c = \left(\frac{S(T)}{\rho}\right)_{c,s} + \left(\frac{S(T)}{\rho}\right)_{c,h}. \quad (4)$$

To derive these, consider energy transfers T' in collisions. The mass collision stopping power integrates differential cross-sections:

$$\left(\frac{S(T)}{\rho}\right)_c = \int_{T'_{\min}}^{T'_{\max}} T' dQ_c(T'), \quad (5)$$

where $dQ_c(T')$ is the differential mass collision coefficient (cm^2/g per MeV or m^2/kg per J). An arbitrary boundary H separates soft ($T' < H$) and hard ($T' > H$) terms:

$$\left(\frac{S(T)}{\rho}\right)_c = \int_{T'_{\min}}^H T' dQ_{c,s}(T') + \int_H^{T'_{\max}} T' dQ_{c,h}(T'). \quad (6)$$

1.2.2 Maximum Energy Transfer

For a heavy particle (mass $M \gg m_e$) colliding with a free electron:

$$T'_{\max} = \frac{2m_e c^2 \beta^2 \gamma^2}{1 + 2\gamma(m_e/M) + (m_e/M)^2} \approx 2m_e c^2 \beta^2 \gamma^2, \quad (7)$$

where $\beta = v/c$, $\gamma = 1/\sqrt{1 - \beta^2}$. For non-relativistic limit ($\gamma \approx 1$): $T'_{\max} \approx 2m_e v^2 = 4(m_e/M)T$ (since $T = (1/2)Mv^2$).

Derivation (kinematics): In the lab frame, for head-on collision, conservation of energy and momentum yields the above. For electrons incident on electrons: $T'_{\max} = T/2$ (indistinguishable particles). For positrons: $T'_{\max} = T$ if no annihilation.

The minimum $T'_{\min}$ relates to the mean excitation potential I :

$$T'_{\min} \approx \frac{I^2}{T'_{\max}} \approx \frac{I^2}{2m_e c^2 \beta^2 \gamma^2}, \quad (8)$$

as $T'_{\min} T'_{\max} \approx I^2$ from quantum considerations (Bethe's approximation).

1.3 Bethe-Bloch Formula (Heavy Particles)

For heavy ions ($M \gg m_e$, charge ze), the collision mass stopping power is the Bethe-Bloch formula. We derive it step by step from classical and quantum considerations, as outlined in the documents and standard derivations (e.g., Bohr's classical model extended by Bethe).

1.3.1 Classical Derivation (Bohr's Model)

Consider a heavy particle (charge ze , velocity $v = \beta c$) passing an atom at impact parameter b . The impulse on a bound electron (charge $-e$) is from the transverse electric field.

The force on the electron: $F_{\perp} = \frac{ze^2}{4\pi\epsilon_0} \frac{b}{(b^2 + (vt)^2)^{3/2}}$ (Coulomb, non-relativistic).

The momentum transfer: $\Delta p_y = \int_{-\infty}^{\infty} F_{\perp} dt = \frac{ze^2}{4\pi\epsilon_0} \frac{2}{bv}$ (integrate over time t).

Energy transfer: $\Delta E = \frac{(\Delta p_y)^2}{2m_e} = \frac{2z^2 e^4}{(4\pi\epsilon_0)^2 m_e v^2 b^2}$.

For many electrons, integrate over impact parameters. Number of collisions in dx with b to $b + db$: $dN = 2\pi b db \cdot n_e dx$, where $n_e = \rho N_A Z/A$ (electrons per volume).

Total energy loss: $-dE/dx = \int_{b_{\min}}^{b_{\max}} \Delta E dN/dx = \frac{2\pi z^2 e^4 n_e}{(4\pi\epsilon_0)^2 m_e v^2} \ln\left(\frac{b_{\max}}{b_{\min}}\right)$.

$b_{\max} \approx v/\omega$ (adiabatic limit, ω mean orbital frequency $\approx I/\hbar$). $b_{\min} \approx ze^2/(4\pi\epsilon_0 m_e v^2)$ (classical minimum for $\Delta E \approx m_e v^2/2$).

Thus: $-dE/dx \approx \frac{4\pi z^2 e^4 n_e}{(4\pi\epsilon_0)^2 m_e v^2} \ln\left(\frac{2m_e v^2}{I(1 - \beta^2)} - \beta^2\right)$ (relativistic adjustments).

In cgs units: $C = \pi(N_A Z/A)r_0^2 = 0.150Z/A \text{ cm}^2/\text{g}$, $r_0 = e^2/(m_e c^2) = 2.818 \times 10^{-13} \text{ cm}$.

1.3.2 Quantum Derivation (Bethe-Bloch)

Bethe used Born approximation ($v \gg$ atomic electron velocities).

The differential cross-section for energy transfer T' : $d\sigma/dT' = \frac{2\pi z^2 e^4}{(4\pi\epsilon_0)^2 m_e v^2 T'^2}$ (Rutherford-like for free electrons).

Integrating: $-dT/dx = n_e \int_{T'_{\min}}^{T'_{\max}} T' (d\sigma/dT') dT' = \frac{2\pi z^2 e^4 n_e}{(4\pi\epsilon_0)^2 m_e v^2} \ln\left(\frac{T'_{\max}}{T'_{\min}}\right)$.

With $T'_{\max} \approx 2m_e v^2$, $T'_{\min} \approx I^2/(2m_e v^2)$: $\ln(2m_e v^2/I) - \ln(1 - \beta^2)/2 - \beta^2/2$ (relativistic).

Full relativistic Bethe-Bloch:

$$-\frac{1}{\rho} \frac{dT}{dx} = \frac{4\pi N_A r_e^2 m_e c^2 z^2}{A\beta^2} Z \left[\ln\left(\frac{2m_e c^2 \beta^2 \gamma^2}{I(1 - \beta^2)}\right) - \beta^2 - \frac{C}{Z} - \frac{\delta}{2} \right], \quad (9)$$

where $r_e = e^2/(4\pi\epsilon_0 m_e c^2)$ (SI units).

1.3.3 Soft-Collision Term

For soft collisions ($b \gg a$, $T' < H$):

$$\left(\frac{dT}{\rho dx}\right)_{c,s} = \frac{2Cm_0 c^2 z^2}{\beta^2} \left[\ln\left(\frac{2m_0 c^2 \beta^2 H}{I^2(1 - \beta^2)}\right) - \beta^2 \right], \quad (10)$$

with $C = 0.150Z/A$, $m_0 = m_e$.

Verified with cyclotron protons.

1.3.4 Hard-Collision Term

Define $K = 0.307075 \text{ MeV cm}^2/\text{g}$ and $r_e = e^2/(4\pi\epsilon_0 m_e c^2)$. For ions/protons of charge ze traversing a material with atomic number Z and mass number A :

$$-\frac{1}{\rho} \frac{dE}{dx} = K \frac{Z}{A} \frac{z^2}{\beta^2} \left[\frac{1}{2} \ln\left(\frac{2m_e c^2 \beta^2 \gamma^2 T'_{\max}}{I^2}\right) - \beta^2 - \frac{\delta(\beta\gamma)}{2} - \frac{C(Z, \beta)}{Z} \right]. \quad (11)$$

Here δ is the density-effect correction (medium polarization at high γ) and C/Z the shell correction (inner shells; important at low β). The $1/\beta^2$ rise as particles slow is the origin of the Bragg peak.

1.4 Corrections

1.4.1 Shell Correction

At low projectile velocities, where v becomes comparable to the orbital velocities of bound atomic electrons, the assumptions underlying the first Born approximation in the Bethe–Bloch theory break down. In this regime, the stopping power is overestimated because the atomic electrons are no longer quasi-free but instead bound in discrete shells (K, L, M, etc.).

To account for this, a *shell correction* term C/Z is introduced, which subtracts the spurious contributions from tightly bound electrons:

- C/Z depends on the projectile velocity β but not on its charge z ; thus, all particles with the same β experience the same shell correction.
- The correction is significant only in the low- β region ($\beta \lesssim 0.1$), i.e. near the end of the particle range where the Bragg peak occurs.
- For typical therapeutic proton and ion beams (energies of tens to hundreds of MeV/nucleon), C/Z is non-negligible in the last few millimeters of range and therefore impacts the exact shape and height of the Bragg peak.

Analytical estimates for C/Z were developed by Bloch, later refined by Barkas and Sternheimer. In practice, C/Z values are obtained from experimental data or tabulated parameterizations (e.g. ICRU reports). A compact approximation often used is

$$\frac{C}{Z} \approx \frac{0.42237}{\beta^2} + \frac{0.0304}{\beta^4} - \frac{0.0003814}{\beta^6},$$

which reproduces the dominant contribution of K- and L-shell electrons at low β .

1.4.2 Density Effect Correction

The functional form of $\delta(\beta\gamma)$ is material-dependent. Following Sternheimer's parameterization, it can be expressed piecewise as

$$\delta(\beta\gamma) = \begin{cases} \delta_0 10^{2(\log_{10} \beta\gamma - x_0)} & \beta\gamma < x_0, \\ 2 \ln(10) (\log_{10} \beta\gamma - x_0) + a (x_1 - \log_{10} \beta\gamma)^m & x_0 \leq \log_{10} \beta\gamma < x_1, \\ 2 \ln(10) (\log_{10} \beta\gamma - C) & \log_{10} \beta\gamma \geq x_1, \end{cases}$$

where the parameters $x_0, x_1, C, a, m, \delta_0$ depend on the medium (empirically tabulated).

A simplified asymptotic expression (valid only at very high γ) is

$$\delta \approx 2 \ln(\gamma) + \ln\left(\frac{\hbar\omega_p}{I}\right) - 1,$$

where $\hbar\omega_p = 28.816 \sqrt{\rho Z/A}$ eV is the plasma energy of the medium. This form illustrates that δ grows logarithmically with γ and eventually compensates the $\ln(\gamma^2)$ rise, leading to a saturation of the stopping power (the “Fermi plateau”).

Accounts for overestimation when β low.

1.5 Multiple Coulomb Scattering (MCS)

The projected RMS angle for thickness x is (Molière's theory):

$$\theta_0 \simeq \frac{13.6 \text{ MeV}}{\beta pc} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln\left(\frac{x}{X_0}\right) \right], \quad (12)$$

with radiation length X_0 and momentum p . MCS increases lateral penumbra with depth.

1.6 Radiation Yield

Fraction of initial kinetic energy T_0 emitted as radiation:

$$Y(T_0) = \frac{E_{\text{rad}}}{T_0} = \int_0^{T_0} \frac{(S_r/\rho)}{(S/\rho)} \frac{dT}{T_0}. \quad (13)$$

For heavy ions, $Y \approx 0$ (negligible bremsstrahlung). For electrons, $S_r \propto Z^2 T/m_e^2$, so significant at high T.

1.7 Range of Charged Particles

1.7.1 Definition

Continuous slowing-down approximation (CSDA) range:

$$R_{\text{CSDA}}(T_0) = \int_0^{T_0} \frac{dT}{S(T)} = \rho \int_0^{T_0} \frac{dT}{\rho S(T)}. \quad (14)$$

Assumes straight path, neglects straggling.

1.7.2 Range Straggling

The CSDA range assumes a continuous and deterministic energy loss along the path, but in reality the stochastic nature of individual collisions leads to fluctuations in the energy deposition, known as *energy straggling*. As a result, particles of the same initial energy T_0 exhibit a spread in their actual ranges.

Thin absorber: Landau distribution. For sufficiently thin absorbers, where the thickness is small compared to the mean free path of hard collisions, the energy-loss probability distribution is highly asymmetric with a long high-energy tail. This is described by the *Landau distribution*,

$$f(\lambda) = \frac{1}{\pi} \int_0^\infty \exp[-t \ln t - \lambda t] \sin(\pi t) dt,$$

where λ is the reduced variable depending on the thickness and mean energy loss. The Landau distribution captures the enhanced probability of rare, large-energy transfers (“ δ -rays”). Its most probable value (MPV) is less than the mean.

Thicker absorbers: Vavilov and Symon theory. As absorber thickness increases, multiple collisions contribute and the central limit theorem drives the distribution towards a Gaussian. The intermediate regime is well described by the *Vavilov distribution*, which interpolates between Landau-like behavior and a Gaussian limit depending on the parameter $\kappa = \xi/T_{\max}$, with ξ the mean energy loss in the layer. For $\kappa \ll 1$ the Landau form is recovered, while for $\kappa \gg 1$ a Gaussian emerges.

Symon (1941) provided a theoretical framework for the energy straggling variance in thick absorbers. The variance of the energy loss σ^2 can be approximated as

$$\sigma^2 \approx 4\pi N_A r_e^2 m_e c^2 z^2 \frac{Z}{A} \rho x,$$

which scales linearly with path length x . In the *thick-target* limit, the resulting range distribution of charged particles approaches a Gaussian centered at the CSDA range, with a standard deviation $\sigma_R \sim (1\%-2\%) R_{\text{CSDA}}$ for protons in water at therapeutic energies.

In summary:

- **Thin layers:** asymmetric Landau distribution, dominated by rare hard collisions.
- **Intermediate thickness:** Vavilov distribution interpolating between Landau and Gaussian.
- **Thick layers:** Gaussian limit described by Symon's theory, variance $\propto x$.

These straggling effects contribute to the finite width of the Bragg peak and are crucial for accurate range prediction in clinical particle therapy.

1.7.3 Heavy Particles

Range scales as M/z^2 (since $S \propto z^2/M$ via $\beta^2 \propto T/M$). Bragg peak: $S \propto 1/\beta^2$, peaks at low β .

1.7.4 Electrons

Multiple scattering and straggling make range stochastic. Use projected range (along initial direction) or extrapolated range. No sharp Bragg peak; dose builds to maximum at $\sim R/2$.

1.8 Dose Calculations

Absorbed dose $D = d\epsilon/dm$ (energy per mass):

$$D = \frac{1}{\rho} \int \Phi(T) \left(-\frac{dT}{dx} \right)_c dT, \quad (15)$$

for fluence $\Phi(T)$.

1.8.1 Heavy Particles

For monoenergetic beam through foil thickness $t < R$:

$$D \approx \frac{\Phi}{\rho} \frac{(T_0 - T_{\text{ex}})}{t}, \quad (16)$$

where T_{ex} is exit energy. Bragg curve shows peak near end.

1.8.2 Electrons

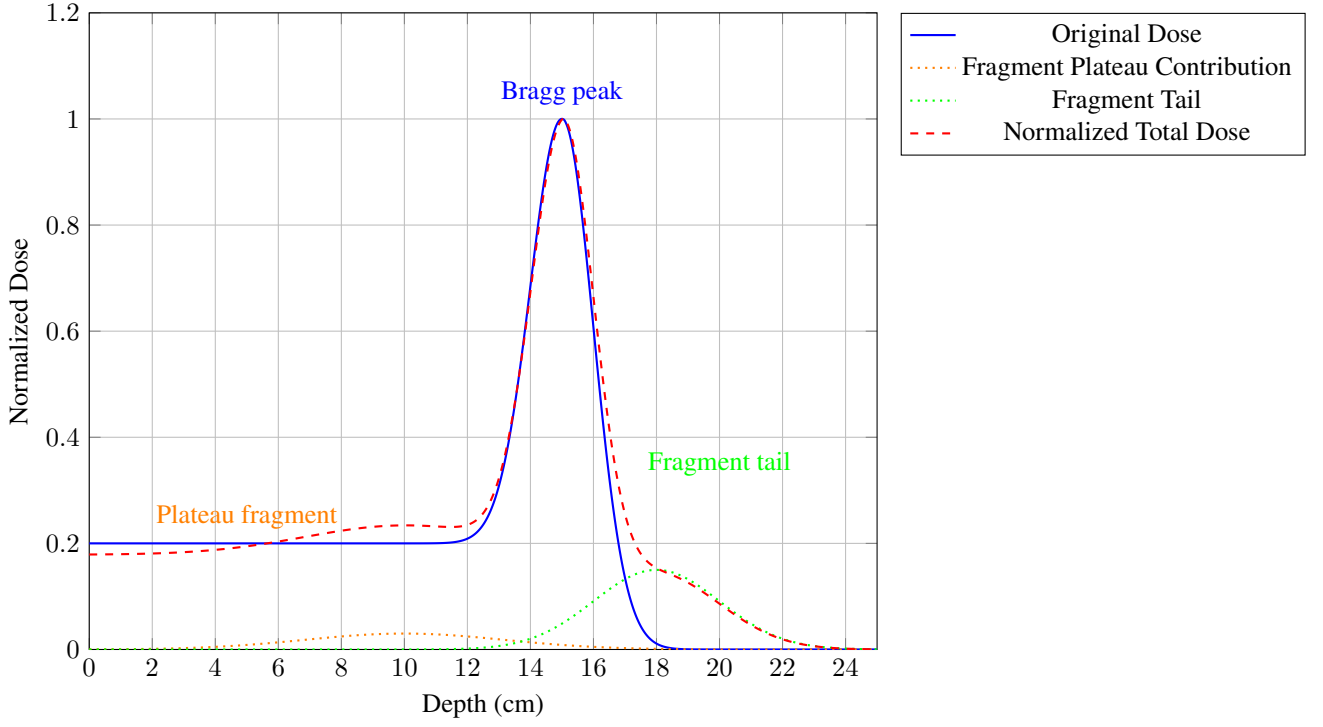
Include radiation loss:

$$D \approx \frac{\Phi}{\rho t} [T_0(1 - Y(T_0)) - T_{\text{ex}}(1 - Y(T_{\text{ex}}))]. \quad (17)$$

Diffuse dose profile due to scattering.

1.9 Bragg Curve

For heavy ions: Sharp dose peak just before stopping (inverse v^2 dependence). Electrons: Gradual rise to broad maximum, then fall-off. Nuclear interactions add minor tail for ions.



The plot shows the normalized dose distribution for heavy ions. The *Original Dose* (blue) exhibits a sharp Bragg peak near the end of the particle range due to the inverse v^2 dependence. The *Fragment Plateau Contribution* (orange) represents low-dose deposition from secondary fragments in the plateau region. The *Fragment Tail* (green) shows a minor dose beyond the Bragg peak due to nuclear interactions. The *Normalized Total Dose* (red dashed) combines all components, giving a continuous and normalized representation of the dose distribution.

2 Beam Delivery in Proton Therapy and Heavy Ion Therapy

Beam delivery systems in proton therapy and heavy ion therapy are designed to precisely deposit ionizing radiation dose within the target tumor volume while minimizing exposure to surrounding healthy tissues. These systems leverage the unique physical properties of charged particles, such as protons (mass $m_p \approx 1.67 \times 10^{-27}$ kg, charge $+e$) and heavier ions (e.g., carbon-12 ions, mass $\approx 12m_p$, charge up to $+6e$), which exhibit a finite range in matter and a characteristic Bragg peak—a sharp increase in energy deposition near the end of their trajectory. This contrasts with photon-based therapies (e.g., X-rays), where dose decreases exponentially with depth.

The primary goal is to achieve a *spread-out Bragg peak (SOBP)* that conforms to the tumor's depth extension, typically on the order of

$$1\text{--}30 \text{ cm in tissue-equivalent depth (g/cm}^2\text{)}.$$

Beam delivery methods can be broadly classified into

- **Passive scattering:** broadening the beam laterally and modulating depth passively,
- **Active scanning:** dynamically steering narrow beams.

Heavy ion therapy (e.g., carbon ions) follows similar principles but benefits from higher linear energy transfer (LET), typically

$$10\text{--}100 \text{ keV}/\mu\text{m} \quad \text{vs.} \quad \sim 0.2 \text{ keV}/\mu\text{m} \text{ for protons,}$$

leading to denser ionization and enhanced biological effectiveness, with

$$\text{RBE} \approx 2\text{--}3 \text{ for carbon ions} \quad \text{vs.} \quad \sim 1.1 \text{ for protons.}$$

Key concepts include:

- **Range:** The depth at which particles stop, scaling as $R \propto E^{1.75-2}/(Z^2 A)$, where E is energy, Z is atomic number, and A is mass number. For 200 MeV protons, $R \approx 25$ cm in water; for 400 MeV/u carbon ions, $R \approx 20$ cm.

- **Lateral penumbra:** The beam's edge sharpness, typically 2–10 mm (80%–20% dose fall-off), broader for deeper ranges due to multiple Coulomb scattering.

- **Dose uniformity:** Required to be within $\pm 2\text{--}5\%$ across the field.

Below, we organize the knowledge points hierarchically, focusing on proton therapy while noting extensions to heavy ions.

2.1 Physical Principles of Charged Particle Interactions in Matter

Understanding beam delivery requires the physics of particle-matter interactions, governed by electromagnetic and nuclear processes.

2.1.1 Energy Loss Mechanisms and the Bethe-Bloch Formula

Charged particles lose energy primarily through inelastic collisions with atomic electrons (collisional loss) and, to a lesser extent, nuclear interactions. The mean energy loss rate (stopping power) is described by the Bethe-Bloch equation:

$$-\frac{dE}{dx} = \frac{4\pi z^2 e^4 N_A \rho Z}{m_e c^2 \beta^2 A} \left[\ln \left(\frac{2m_e c^2 \beta^2 \gamma^2}{I(1 - \beta^2)} \right) - \beta^2 - \frac{\delta}{2} \right],$$

Radiative losses (bremsstrahlung) are negligible for protons below 500 MeV but relevant for electrons.

2.1.2 Range and Range Straggling

The range R is the integral of the inverse stopping power:

$$R(E_0) = \int_0^{E_0} \frac{dE}{-dE/dx}.$$

For practical calculations, empirical approximations are used, e.g., for protons in water: $R \approx (E/\text{MeV})^{1.77}/10$ cm (accurate to 1% for 50–250 MeV).

Range straggling: Due to stochastic energy losses, the range spreads Gaussian-like with standard deviation

$$\sigma_R \approx \sqrt{\int_0^R (d\sigma_E^2/dx) dx},$$

where $d\sigma_E^2/dx$ is the energy straggling variance (Bohr theory: $\sigma_E^2 \propto 4\pi z^2 e^4 N Z x$). For protons, $\sigma_R/R \approx 1\%$ (e.g., $\sigma_R \approx 0.2\text{--}0.3$ cm for $R = 25$ cm), broadening the distal fall-off to 1–5 mm (80%–20%).

For heavy ions, nuclear reactions cause additional fragmentation, extending the dose tail beyond the range by 1–5% of R .

2.1.3 Multiple Coulomb Scattering and Lateral Beam Broadening

Lateral deflection follows Molière theory, with angular spread $\theta_{rms} \approx (13.6 \text{ MeV}/(\beta pc)) \sqrt{x/X_0} (1 + 0.038 \ln(x/X_0))$, where p is momentum, x is depth, and X_0 is radiation length (36 cm for water). This yields a lateral penumbra of 2–5 mm for shallow depths, increasing to 5–10 mm at 30 cm.

2.2 Passive Beam Delivery Techniques

Passive methods use scattering foils to broaden the beam laterally and range modulators for depth spreading, suitable for uniform fields up to $30 \times 40 \text{ cm}^2$.

2.2.1 Single Scattering

Objective: Uniform dose for small fields (diameter < 7 cm), ranges 3.5–30 g/cm², modulation up to 20 g/cm².

Setup: A fixed scatterer (high-Z material like lead for efficient scattering, low-Z like plastic for minimal energy loss) broadens the pencil beam. A range modulator wheel (propeller-like with varying thickness) creates the SOBP by stacking shifted Bragg peaks.

Physics: The modulator wheel rotates at 600 RPM, varying thickness $t(\phi)$ to shift energy as $\Delta E \propto \rho t \cdot dE/dx$. Beam gating adjusts intensity for flat SOBP. Dose profile: Plateau dose 80–90% of peak, with 1–5 mm distal fall-off.

Limitations: Inefficient for large fields (low transmission 10–20%), higher neutron production (order of 10^{-3} – 10^{-2} n/proton).

2.2.2 Double Scattering

Objective: Uniform dose for larger fields (diameter < 24 cm for ranges 4–25 g/cm²; < 12 cm for 4–28 g/cm²), full modulation.

Setup: First scatterer (fixed) for initial broadening, range modulator, second scatterer (contoured high-Z/low-Z for uniform fluence), patient-specific collimator (brass block) and range compensator (plastic bolus shaped to distal tumor surface).

Physics: The second scatterer optimizes Gaussian beam profiles for flatness. Relative dose profiles show uniformity within $\pm 2\%$. Neutron dose higher than scanning (10-fold).

Advantages: Simple, robust;

Disadvantages: Requires patient-specific hardware, beam loss 50–70%.

For heavy ions, double scattering is less common due to sharper intrinsic peaks but similar principles apply.

2.3 Active Beam Delivery Techniques

Active methods dynamically scan narrow beams (spot size $\sigma \approx 3\text{--}10\text{ mm}$), enabling intensity-modulated proton therapy (IMPT) without scatterers.

2.3.1 Uniform Scanning (Wobbling)

Objective: Uniform dose for large fields (up to $30 \times 40\text{ cm}^2$), ranges $4\text{--}32\text{ g/cm}^2$, full modulation.

Setup: Fixed scatterer for mild broadening, scanning magnets for X-Y sweeping (saw-tooth pattern at 3 Hz), stepping range modulator for layers.

Physics: Each "mini-irradiation" treats an iso-energy slice; beam paused between layers for absorber adjustment. Frame time 333 ms . Dose built by summing layers: $D(z) = \sum_i w_i D_{\text{Bragg},i}(z)$, with weights w_i for flat SOBP.

Orders of magnitude: Irradiation time $1\text{--}2\text{ min/field}$; lateral penumbra $2\text{--}4\text{ mm}$.

2.3.2 Pencil Beam Scanning (PBS)

Objective: Conformal dose for arbitrary targets (up to $30 \times 40\text{ cm}^2$), ranges $5\text{--}32\text{ g/cm}^2$, modulation $<28\text{ g/cm}^2$, enabling IMPT.

Setup: Scanning magnets steer pencil beams in X-Y, energy varied layer-by-layer (via accelerator or degrader). No patient-specific hardware needed.

Physics: Tumor sliced into iso-energy layers; each spot delivers dose $D(\mathbf{r}) \propto \int \phi(E)(-dE/dx)dE$, optimized via inverse planning. Multiple repaints ($10\text{--}20$) mitigate motion interplay. Spot spacing $\sigma/2$ for uniformity.

Derivation of dose conformity: In IMPT, fluence $\phi(\mathbf{r}_\perp, E)$ is optimized to minimize $\sum (\hat{D} - D_{\text{prescribed}})^2 + \lambda \cdot \text{constraints}$. RBE-weighted dose for heavy ions: $D_{\text{bio}} = \text{RBE} \cdot D_{\text{phys}}$, with RBE models (e.g., local effect model) accounting for LET dependence.

Advantages: Best conformity (+++), lower activation/neutrons;

Disadvantages: Sensitive to motion, longer times ($1\text{--}2\text{ min}$).

For heavy ions, PBS exploits higher RBE at the Bragg peak for enhanced tumor kill.

2.4 Future Directions and Challenges

Integration of real-time imaging (e.g., MRI-guided) and adaptive planning to handle range uncertainties ($\pm 3\text{--}5\text{ mm}$). Research into ultra-high dose rates (FLASH therapy, $>40\text{ Gy/s}$) may reduce normal tissue toxicity. For heavy ions, helium or oxygen beams could balance proton-like sharpness with carbon-like RBE.

3 Introduction to Radiobiology

3.1 History of Radiation Injuries in Humans

- 1895: Roentgen discovered x-rays.
- 1896: Becquerel discovered radioactivity.
- 1896–1897: Freund demonstrated disappearance of a hairy mole with x-rays.
- 1898: Curies isolated radium.
- 1901: Becquerel and Pierre Curie experienced radiation burns from radium in vest pocket.
- 1915: British Roentgen Society issued guidelines for x-ray protection.
- 1920s: Coutard developed fractionated radiation therapy.
- 1897–1960: **Ringworm treatment:** children received $3\text{--}6\text{ Gy}$ to scalp $\Rightarrow$ acute nausea, epilation; late effects permanent alopecia, skin/brain/thyroid cancers.
- 1917–1926: **Radium Girls:** dial painters ingested radium $\Rightarrow$ bone incorporation, dental problems, deaths, workplace safety issues.
- 1945: Atomic bombs in Hiroshima/Nagasaki, doses $4.5\text{--}8.7\text{ Gy}$ at 1 km $\Rightarrow$ acute poisoning, burns; late effects leukemia, cancers.
- 1986: Chernobyl accident: 137 with ARS, 28 deaths, 6000 thyroid cancers, ~ 9000 radiation-associated deaths expected; cardiovascular complications reported.

3.2 Research Subjects in Radiobiology

- **Humans:** historical events, patient data; real but heterogeneous, ethical limits.
- **Animals:** controlled environment, genetic modification possible; limited by animal ethics.
- **Cells in culture:** isolate single type, cycle synchronization; simplified, low ethical burden.

3.3 Units in Radiobiology

Quantity	Unit	Definition / Conversion
Absorbed dose	Gray (Gy)	1 Gy = 1 J/kg
	rad	1 rad = 0.01 Gy $\Rightarrow$ 1 Gy = 100 rad
Equivalent dose	Sievert (Sv)	Sv = Gy $\times$ Q (quality factor)
	rem	1 rem = 0.01 Sv $\Rightarrow$ 1 Sv = 100 rem
Radioactivity	Becquerel (Bq)	1 Bq = 1 disintegration/s
	Curie (Ci)	1 Ci = 3.7×10^{10} Bq $\Rightarrow$ 1 Bq = 2.703×10^{-11} Ci
Exposure	Roentgen (R)	1 R $\approx 2.58 \times 10^{-4}$ C/kg $\approx$ 0.01 Sv (for X- or gamma rays in air)

Table 1: Radiation units with conversion formulas.

3.4 Background Radiation

- Australia average: ~ 1.7 mSv/year.
- Global range: 1–13 mSv/year.

4 Radiation Effects on Biological Systems

4.1 Acute and Late Effects

- **Acute (days–months):** GI (nausea, vomiting, diarrhea), skin (erythema, desquamation), hematopoietic (lymphopenia, infection), hair loss.
- **Late (months–years):** cancers, fibrosis, vascular damage, cognitive/endocrine impairment, infertility.
- First response: DNA damage response (DDR), inflammation, cytokines, immune, psychological fatigue.

4.2 Radiation Syndromes and Corresponding Dose Ranges

Acute high-dose radiation exposure can lead to distinct clinical syndromes depending on the absorbed dose:

- **Hematopoietic Syndrome (Bone Marrow Syndrome):**
 - Dose: ~ 1 –8 Gy (whole-body exposure, low LET).
 - Onset: Days to weeks post-exposure.
 - Pathophysiology: Bone marrow suppression $\rightarrow$ leukopenia, thrombocytopenia, anemia.
 - Clinical features: Infection susceptibility, bleeding, fatigue.
 - Mortality: 50% LD₅₀ at ~ 4 –5 Gy without medical support.
- **Gastrointestinal (GI) Syndrome:**
 - Dose: ~ 8 –30 Gy.
 - Onset: Hours to days post-exposure.
 - Pathophysiology: Destruction of intestinal crypt stem cells.
 - Clinical features: Severe nausea, vomiting, diarrhea, dehydration, electrolyte imbalance.
 - Mortality: Often fatal within 1–2 weeks without intensive supportive care.
- **Cerebrovascular/Central Nervous System (CNS) Syndrome:**
 - Dose: > 30 Gy.
 - Onset: Minutes to hours post-exposure.

- Pathophysiology: Massive vascular and neural tissue damage.
- Clinical features: Confusion, ataxia, seizures, loss of consciousness, edema, intracranial pressure rise.
- Mortality: Usually fatal within 48 hours.

Note: These syndromes are typically observed after acute, whole-body exposure. Partial-body exposure or fractionated doses may lead to modified presentations.

4.3 Free Radical Formation and DNA Damage

- **Direct effect:** ~30% DNA damage (low LET); higher with high LET.
- **Indirect effect:** ~70% DNA damage (low LET) via water radiolysis → ROS.
- Damage types: base oxidation, abasic sites, SSBs, DSBs.
- Typical yield: 20–40 DSB/Gy.
- Oxygen fixation: permanent damage.
- Repair: ~70% correct for x-ray-induced DSBs.

4.4 Target Theory and Cell Survival

- Cell death = loss of reproductive capacity (clonogenic assay).
- Survival fraction: $S(D) = \exp(-\alpha D - \beta D^2)$.
- α/β ratio: indicator of tissue response (low $\alpha/\beta \sim 3$ Gy late-responding tissue; high ~ 10 Gy tumours).
- RBE: $\frac{D_{\text{reference}}}{D_{\text{test}}}$ at same biological effect.
- OER: $\frac{D_{\text{anoxic}}}{D_{\text{oxic}}}$, typically 2–3 for low LET.

4.5 Mechanisms of Radiation Effects and Immunology

Radiation affects cells and tissues through a combination of direct DNA damage, indirect effects via free radicals, and modulation of tissue and immune responses. Key concepts include:

- **The 5 Rs of Radiobiology:** Fundamental principles governing tissue response to fractionated radiotherapy:
 - **Repair:** Sublethal DNA damage can be repaired between fractions; efficiency depends on cell type, cell cycle phase, and DNA repair capacity.
 - **Redistribution:** Cells move through the cell cycle between fractions; sensitivity varies by phase (e.g., M and G2 most sensitive, S least sensitive).
 - **Repopulation:** Surviving cells can proliferate between fractions, potentially reducing therapeutic effect; tumor and normal tissue repopulation rates differ.
 - **Reoxygenation:** Hypoxic cells are radioresistant; oxygen diffusion during intervals increases radiosensitivity of previously hypoxic regions.
 - **Radiosensitivity:** Intrinsic susceptibility of cells/tissues; influenced by genetic factors, microenvironment, and cell differentiation status.
- **Deterministic Effects:** These effects exhibit a threshold dose below which they do not occur. Severity increases with dose:
 - Examples: skin erythema, cataracts, fibrosis, organ atrophy.
 - Mechanism: Occurs when a critical fraction of cells in a tissue are lethally damaged, impairing organ function.
 - Clinical relevance: Predictable, dose-limiting side effects in radiotherapy.
- **Stochastic Effects:** These effects have no threshold; the probability increases with dose, but severity is independent of dose:
 - Examples: radiation-induced carcinogenesis, genetic mutations.
 - Mechanism: Damage in a single or few cells can lead to tumorigenesis; DNA repair fidelity and misrepair contribute.
 - Clinical relevance: Long-term risk assessment; relevant for low-dose exposures (e.g., occupational or environmental).

- **Therapeutic Ratio (TCP/NTCP):**

- **TCP (Tumor Control Probability):** Probability that all clonogenic tumor cells are eradicated.
- **NTCP (Normal Tissue Complication Probability):** Probability of causing clinically significant normal tissue toxicity.
- **Optimization:** Fractionation schemes and advanced delivery techniques aim to maximize TCP while minimizing NTCP.

- **Radiation-Induced Immune Modulation and Evasion:**

- Radiation can activate immune responses via release of danger signals, DAMPs, and neoantigens, enhancing tumor immunogenicity.
- Tumors can evade immune surveillance post-irradiation by:
 - * **Upregulating inhibitory checkpoints:** PD-L1, CTLA-4 reduce T-cell activity.
 - * **Expressing "don't eat me" signals:** CD47 prevents macrophage phagocytosis.
 - * **Downregulating antigen presentation:** Reduced MHC I expression diminishes cytotoxic T lymphocyte recognition.
- Clinical implications: Combining radiotherapy with immunotherapy (e.g., checkpoint inhibitors) can synergistically improve tumor control.

5 Radiation-Induced DNA Damage and Repair

5.1 DNA Double-Strand Breaks (DSBs) and Repair Mechanisms

Ionizing radiation induces DNA damage via direct ionization and indirect free radical formation. DSBs are the most lethal lesions.

- **DSB sensing and signaling:**

- **ATM and ATR kinases:** Activated by DSBs and replication stress, phosphorylate downstream effectors to initiate DNA damage response (DDR).
- **DNA-PK complex:** Composed of Ku70/80 and DNA-PKcs; recognizes broken ends and recruits NHEJ machinery.
- **MRN complex:** MRE11-RAD50-NBS1; detects DSBs and promotes end resection for HR.
- **H2AX phosphorylation (γ H2AX):** Ser139 phosphorylation marks DSB sites; forms foci visible via immunofluorescence.

- **Homologous Recombination (HR, error-free):**

- Uses sister chromatid as template; active in S/G2 phase.
- Key proteins: RAD51 (strand invasion), BRCA1/2 (recruitment, repair coordination), helicases and polymerases.
- Accurate repair; prevents mutations and chromosomal rearrangements.

- **Non-Homologous End Joining (NHEJ, error-prone):**

- Ligates DSB ends without a template; active throughout cell cycle, predominant in G1.
- Key proteins: Ku70/80 (end recognition), DNA-PKcs (kinase activity), LIG4/XRCC4 (ligation).
- Faster but may introduce small insertions/deletions, potentially mutagenic.

5.2 Radiation-Induced Cell Death Pathways

Radiation induces multiple cell death mechanisms depending on dose, cell type, and cell cycle stage:

- **Mitotic catastrophe:** Premature or inappropriate mitosis due to unrepaired DSBs; leads to micronuclei or multinucleated cells; common in dividing cells after IR.
- **Apoptosis:** Programmed cell death; features include chromatin condensation, DNA fragmentation, membrane blebbing; mediated by p53, caspases; prominent in lymphocytes, spermatocytes, thymocytes.
- **Necrosis:** Passive death from overwhelming damage; cell swelling, membrane rupture, inflammation; typical of high-ablative doses.
- **Senescence:** Permanent growth arrest, β -gal positive; involves p16/p21/p53; contributes to late tissue effects.
- **Autophagy:** Lysosome-mediated degradation of organelles; can promote survival or cell death depending on context; type II programmed cell death.

5.3 Cell Cycle-Dependent Radiosensitivity

Radiosensitivity varies across cell cycle phases:

- **Order of sensitivity:** late G1 < early S < late S < early G2 < M (most sensitive).
- **Mechanistic basis:**
 - M phase: compact chromosomes, limited repair, high lethality.
 - S phase: active replication; early S has some protection, late S enriched in HR repair capacity.
- **Bergonie–Tribondeau law (1906):** Cells that are undifferentiated, actively proliferating, or metabolically active are more radiosensitive.

5.4 Assays for DNA Damage Detection and Quantification

- **γ H2AX foci assay:** Immunofluorescence detects single DSB foci; high sensitivity (1 DSB/cell); can monitor repair kinetics.
- **Comet assay:** Single-cell gel electrophoresis; alkaline variant detects SSBs and DSBs; tail moment quantifies damage.
- **Micronucleus assay:** Detects chromosomal fragments or mis-segregated chromosomes in binucleated cells; indicates genomic instability.
- **Chromosomal aberration analysis:** Dicentrics, rings, translocations; used for biodosimetry and mutagenesis studies.
- **Pulse-field gel electrophoresis (PFGE):** Resolves large DNA fragments; detects extensive DSBs in populations; allows assessment of repair kinetics.
- **Plasmid-based in vitro assays:** Circular plasmids exposed to IR; supercoiled/open/linear forms analyzed to quantify SSBs/DSBs and repair efficiency.

6 Relative Biological Effectiveness (RBE)

6.1 Linear-Quadratic Model

The linear-quadratic (LQ) model provides a widely used mathematical framework to describe cell survival following radiation exposure:

$$S(D) = \exp(-\alpha D - \beta D^2),$$

where α represents the probability of lethal damage from a single-track event (linear term), and β represents the contribution from two-track interactions (quadratic term).

For a given survival fraction S , one can solve for the required dose D :

$$\ln(S) = -(\alpha D + \beta D^2),$$
$$D = \frac{-\alpha + \sqrt{\alpha^2 - 4\beta \ln(S)}}{2\beta}.$$

The Relative Biological Effectiveness (RBE) at a specified survival level S is then defined as the ratio of the reference photon dose to the test radiation dose:

$$\text{RBE}(S) = \frac{D_{\text{reference}}(S)}{D_{\text{test}}(S)}.$$

For RBE_{10} , the ratio of D_x and D_i is

$$\frac{D_x}{D_i} = \frac{2\beta_x D_x}{\sqrt{\alpha_i^2 - 4\beta_i \ln 0.1} - \alpha_i} \approx \frac{\alpha_i}{\alpha_x}$$

6.1.1 RBE at the low-dose limit

Let radiation A and B be described by the LQ parameters (α_A, β_A) and (α_B, β_B) respectively, and let D_A and D_B be the doses producing the same surviving fraction S . From the LQ model

$$S = \exp(-\alpha_A D_A - \beta_A D_A^2) = \exp(-\alpha_B D_B - \beta_B D_B^2),$$

hence

$$\alpha_A D_A + \beta_A D_A^2 = \alpha_B D_B + \beta_B D_B^2. \quad (1)$$

The Relative Biological Effectiveness (RBE) of A relative to B at this survival is

$$\text{RBE} = \frac{D_A}{D_B}.$$

From (1) we obtain a quadratic equation for D_A :

$$\beta_A D_A^2 + \alpha_A D_A - (\alpha_B D_B + \beta_B D_B^2) = 0,$$

whose physically relevant root (positive for small positive doses) is

$$D_A = \frac{-\alpha_A + \sqrt{\alpha_A^2 + 4\beta_A(\alpha_B D_B + \beta_B D_B^2)}}{2\beta_A}. \quad (2)$$

Therefore

$$\text{RBE}(D_B) = \frac{D_A}{D_B} = \frac{1}{D_B} \cdot \frac{-\alpha_A + \sqrt{\alpha_A^2 + 4\beta_A(\alpha_B D_B + \beta_B D_B^2)}}{2\beta_A}.$$

Special case $\beta_A = \beta_B$ and in the limit $D_B \rightarrow 0$.

$$\text{RBE}(D_B) = \frac{\beta_A \alpha_B}{\beta_B \alpha_A} = \frac{\alpha_B}{\alpha_A}$$

6.2 LET and Typical RBE Values for Different Particles

Radiation	LET (keV/ μm)	Typical RBE	Notes
Photons (X, γ)	0.2	1	Reference radiation
Protons	0.2–2	1.1	Slight LET increase at distal SOBP
Carbon ions	10–100	2–3	High LET, enhanced RBE in tumor core
Alpha particles	50–150	3–7	High LET, dense ionization, short range
Neutrons (fast)	10–200	3–10	Indirect ionization via recoil protons; energy-dependent
Neutrons (thermal)	< 0.5	2–4	Mostly capture reactions; high biological damage per absorbed dose

Table 2: LET and typical RBE values for different radiation types. Neutron RBE varies with energy.

6.3 Mechanistic Basis of RBE Variation

- **High-LET radiation:** Dense ionizations $\rightarrow$ complex, clustered DNA damage $\rightarrow$ harder to repair $\rightarrow$ higher RBE.
- **Low-LET radiation:** Sparse ionizations $\rightarrow$ mostly isolated SSBs/DSBs $\rightarrow$ easier repair $\rightarrow$ lower RBE.
- **Neutrons:** Indirect ionization via recoil nuclei; high LET tracks in tissue $\rightarrow$ complex DSBs $\rightarrow$ higher RBE than photons.
- **Overkill effect:** At extremely high LET, energy deposition exceeds requirement to kill cell; RBE plateaus or decreases.

6.4 Radiobiological Dose Metrics and Fractionation

6.4.1 Surviving Fraction with Fractionation

For n fractions of dose d each, the surviving fraction is described by the linear-quadratic (LQ) model:

$$S = \exp[-n(\alpha d + \beta d^2)].$$

Including accelerated repopulation (with overall treatment time T and kick-off time T_k):

$$S = \exp \left[-n(\alpha d + \beta d^2) + \frac{\ln 2 (T - T_k)}{T_p} \right],$$

where T_p is the potential doubling time.

6.4.2 Biologically Effective Dose (BED)

- **Without repopulation:**

$$\text{BED} = nd \left(1 + \frac{d}{\alpha/\beta} \right)$$

- **With repopulation:**

$$\text{BED} = nd \left(1 + \frac{d}{\alpha/\beta} \right) - \frac{\ln 2 (T - T_k)}{\alpha T_p}$$

- **Adjusted for incomplete repair** (fraction time $T > 10$ h):

$$\text{BED} = RT \left(1 + \frac{KR}{\alpha/\beta} \right), \quad K = \frac{2}{\mu} \left(1 - \frac{1}{\mu T} \right),$$

with repair rate constant $\mu = 0.5 \text{ h}^{-1}$.

6.4.3 Single Fraction Equivalent Dose (SED)

The SED converts a fractionated scheme to an equivalent single fraction dose with 2 Gy/fraction:

$$\text{SED} = nd \frac{1 + d/(\alpha/\beta)}{1 + 2/(\alpha/\beta)}.$$

6.4.4 Tumor Control Probability (TCP)

$$\text{TCP} = \exp \left[-\rho V e^{-\alpha nd} \right],$$

where ρV is the total number of clonogenic cells in the tumor. The required dose for a given TCP can be obtained by:

$$D = \frac{1}{\alpha} \left[\ln(\rho V) - \ln \left(\ln \frac{1}{\text{TCP}} \right) \right].$$

6.4.5 Typical α/β Values

Tissue Type	α/β (Gy)	α	β
Tumor / Early-responding	10	0.35	0.035
Late-responding tissues	3	0.15	0.05

Table 3: Typical α/β values for different tissues.

Notes:

- Larger α/β indicates tissues less sensitive to fractionation (early-responding).
- Smaller α/β indicates tissues more sensitive to fractionation (late-responding).

6.4.6 Clinical Implications

- BED allows comparison of different fractionation schedules.
- RBE-adjusted BED used for particle therapy (protons, carbon ions, neutrons).
- Incorporates effects of fractionation, total dose, repopulation, and repair kinetics.
- TCP model guides dose prescription for achieving tumor eradication probability.

6.5 Clinical Relevance

- **SOBP design:** Spread-out Bragg peak ensures tumor coverage (1–30 cm).
- **Proton therapy:** Low LET, RBE ~ 1.1 ; high conformality minimizes normal tissue dose.
- **Carbon ion therapy:** High LET, RBE 2–3; effective against hypoxic, radioresistant tumors.
- **Neutron therapy:** Limited availability; high RBE can be advantageous for radioresistant tumors but increased normal tissue toxicity.
- RBE is dose-, particle-, tissue-, and endpoint-dependent; must be considered in treatment planning.

7 Microdosimetry

7.1 Definition of microdosimetry

Microdosimetry quantifies the stochastic nature of energy deposition by ionizing radiation in microscopic sites that are biologically relevant (typical site diameters: 1–2 μm ; nanodosimetry: 1–10 nm). It complements macroscopic dosimetry and average quantities such as Linear Energy Transfer (LET) by describing fluctuations, clustering and delta-ray effects that strongly influence biological endpoints (e.g. DNA double-strand breaks).

Key stochastic quantities:

- **Specific energy:** $z = \frac{\epsilon}{m}$ (Gy), where ϵ is the energy imparted to the site and m is the site mass. z is a random variable; the macroscopic absorbed dose is $D = \bar{\epsilon}/m = \mathbb{E}[z]$.
- **Lineal energy:** $y = \frac{\epsilon}{l}$ (keV/ μm), where l is the sampled chord length across the site. y is the stochastic counterpart to LET and depends on site geometry and chord sampling.
- **Single-event spectrum:** $f(z, 1)$ denotes the specific-energy distribution from a single event; the full spectrum $f(z)$ is obtained by compounding over the number of events (track traversals).

7.2 Introduction to microdosimetric concepts

7.2.1 Particle track and inchoate distribution

A charged-particle track comprises the primary particle path plus secondary electrons (delta rays). Energy transfers along the track are commonly modeled as a compound Poisson process:

$$N \sim \text{Poisson}(\bar{n}), \quad Z = \sum_{i=1}^N \epsilon_i,$$

where ϵ_i are iid single-event energy transfers (distribution $f(\epsilon)$). The *inchoate distribution* denotes the unresolved initial spatial energy-deposition pattern before averaging over site geometry; it captures delta-ray excursions that LET neglects.

Limitations of LET:

1. LET is a continuous average dE/dx and does not represent event-to-event fluctuations.
2. Delta-ray spectra and ranges differ between particles of equal LET but different velocities, affecting small-site energy deposition.
3. Track-end and partial-traversal effects change local deposition statistics (important in Bragg peak distal edge).

7.2.2 Event, event size and event site

- **Event:** single energy deposition in a site from primary or secondary charged particle.
- **Event size:** imparted energy ϵ (e.g. in keV).
- **Event site:** chosen geometry (sphere or cylinder) with diameter characteristic of the biological target. Typical masses: for a 1 μm sphere, $m \sim 10^{-15}$ kg; for a 2 nm sphere, $m \sim 10^{-24}$ kg (see numerical examples below).

7.2.3 Specific energy: distributions and moments

Specific energy z distribution $f(z)$ can be decomposed by event-count:

$$f(z) = \sum_{k=0}^{\infty} p(k) f(z; k) \quad \text{with} \quad p(k) = \frac{\bar{n}^k e^{-\bar{n}}}{k!}.$$

Important moments:

$$z_F = \int_0^{\infty} z f(z) dz, \quad z_D = \frac{\int_0^{\infty} z^2 f(z) dz}{\int_0^{\infty} z f(z) dz} = \frac{\langle z^2 \rangle}{\langle z \rangle}.$$

For a compound Poisson process with single-event mean z_1 and variance σ_1^2 , and mean event number $\bar{n}$:

$$\mathbb{E}[Z] = \bar{n} z_1, \quad \text{Var}(Z) = \bar{n} \sigma_1^2 + \bar{n} z_1^2 = \bar{n} z_1^2 (1 + g),$$

where $g \equiv \sigma_1^2 / z_1^2$ is the relative variance of the single-event distribution.

7.2.4 Lineal energy

Lineal energy y is geometry-dependent:

$$y = \frac{\epsilon}{l}, \quad l = \text{chord length},$$

and yields frequency and dose distributions:

$$\bar{y}_F = \int_0^\infty y f(y) dy, \quad d(y) = \frac{y f(y)}{\bar{y}_F}, \quad \bar{y}_D = \int_0^\infty y d(y) dy = \frac{\langle y^2 \rangle}{\langle y \rangle}.$$

For a spherical site of radius R , the mean chord length is $\langle l \rangle = \frac{4R}{3}$ (useful when converting between y and z).

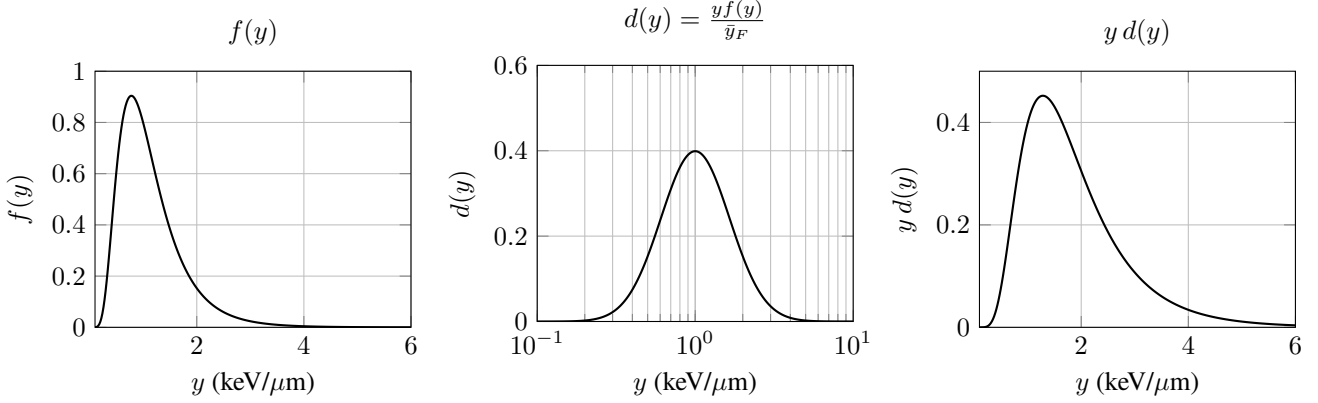


Figure 1: Microdosimetric spectra: only $d(y)$ uses log-x axis.

Typical practical considerations:

- Cutoff or saturation corrections often applied above $y_c \approx 100\text{--}500 \text{ keV}/\mu\text{m}$ to account for overkill.
- Track-averaged, saturation-corrected quantity y^* is frequently used; for $y < 100 \text{ keV}/\mu\text{m}$, $y^* \approx \text{LET}$.

7.3 Multi-event distribution and the Theory of Dual Radiation Action

Multi-event distribution. The statistical distribution of the specific energy z deposited in a microscopic site can be built from the single-event distribution $f_1(z)$. Since events are independent, the number of events ν follows a Poisson distribution with mean n :

$$P(\nu|n) = \frac{e^{-n} n^\nu}{\nu!}, \quad \langle \nu \rangle = n.$$

The multi-event distributions $f_\nu(z)$ are recursively defined by convolution of $f_1(z)$:

$$f_0(z) = \delta(z), \quad f_1(z) = f_1(z), \quad f_2(z) = \int_0^\infty f_1(z') f_1(z - z') dz', \quad \dots$$

$$f_\nu(z) = \int_0^\infty f_1(z') f_{\nu-1}(z - z') dz.$$

The overall probability density of the specific energy is then

$$f(z; n) = \sum_{\nu=0}^\infty P(\nu|n) f_\nu(z) = e^{-n} \sum_{\nu=0}^\infty \frac{n^\nu}{\nu!} f_\nu(z).$$

Thus, $f(z; n)$ is uniquely determined by $f_1(z)$.

Fourier transform method. Using the convolution theorem:

$$F_1(\omega) = \mathcal{F}[f_1(z)], \quad F_\nu(\omega) = (F_1(\omega))^\nu,$$

the generating function becomes

$$F(\omega; n) = \mathcal{F}[f(z; n)] = e^{-n} \sum_{\nu=0}^\infty \frac{(n F_1(\omega))^\nu}{\nu!} = e^{-n(1 - F_1(\omega))}.$$

Hence $f_1(z)$ can be reconstructed by inverting this relation if $f(z; n)$ is known.

Moments of single-event and multi-event distributions. Let $f_1(z)$ be the single-event specific-energy distribution, and $f(z; n)$ the multi-event distribution with average number of events n . Denote the ν -th moment of $f_1(z)$ as

$$m_\nu = \int_0^\infty z^\nu f_1(z) dz,$$

and the ν -th moment of $f(z; n)$ as

$$M_\nu = \int_0^\infty z^\nu f(z; n) dz.$$

Since $f(z; n)$ is the Poisson-weighted sum of convolutions of $f_1(z)$, their moments are related by

$$M_1 = \langle z \rangle = nm_1,$$

$$M_2 = nm_2 + n^2 m_1^2,$$

$$M_3 = nm_3 + 3n^2 m_1 m_2 + n^3 m_1^3, \quad \dots$$

In general, the $(k+1)$ -th single-event moment m_{k+1} can be expressed recursively from multi-event moments M_{k+1} via

$$m_{k+1} = \frac{1}{n} \left[M_{k+1} - \sum_{j=0}^{k-1} \binom{k+1}{j} M_{k-j} m_{j+1} \right], \quad k = 0, 1, 2, \dots$$

Microdosimetric quantities. Using the first and second moments:

$$z_F = m_1 = \text{frequency-mean specific energy}, \quad z_D = \frac{m_2}{m_1} = \text{dose-mean specific energy}.$$

The mean specific energy in the site is

$$\langle z \rangle = M_1 = n z_F,$$

and the second moment of $f(z; n)$ is

$$M_2 = nm_2 + n^2 m_1^2 = n z_F z_D.$$

Physical interpretation.

- m_1 measures the average energy deposition per single event.
- m_2 reflects the variability of single-event depositions.
- M_1 gives the mean total specific energy deposited in the site.
- M_2 encodes the contribution from both single-event variance and Poisson fluctuations of the number of events.

Theory of Dual Radiation Action (TDRA). TDRA interprets biological radiation effects as stochastic outcomes of energy deposition:

- The primary lesions are generated proportionally to the mean specific energy.
- The biological endpoint (e.g. cell death) arises when two lesions interact.
- The probability of interaction depends on their spatial proximity.
- Mathematically, the effect E is proportional to the second moment of z :

$$E \propto \langle z^2 \rangle.$$

In summary, the microdosimetric multi-event formalism links single-event distributions to total specific energy distributions via Poisson statistics and convolution, while TDRA provides a mechanistic hypothesis for translating these stochastic patterns of energy deposition into biological effects.

7.3.1 Mean chord length from Cauchy's surface-area formula

Cauchy's surface-area formula in n -th dimension space is

$$S(K) = \frac{1}{\kappa_{n-1}} \int_{S^{n-1}} V_{n-1}(K \mid \mathbf{u}^\perp) d\mathbf{u},$$

where:

- $S(K)$ denotes the surface area of the convex body $K \subset \mathbb{R}^n$,
- κ_{n-1} is the volume of the unit ball in $\mathbb{R}^{n-1}$,
- $V_{n-1}(K \mid \mathbf{u}^\perp)$ is the $(n-1)$ -dimensional volume of the orthogonal projection of K onto the hyperplane perpendicular to the unit vector $\mathbf{u}$,
- S^{n-1} is the $(n-1)$ -dimensional unit sphere in $\mathbb{R}^n$.

Let K be a convex body in $\mathbb{R}^3$ with volume V and surface area S . For a unit direction $\mathbf{u}$, denote by $A(\mathbf{u})$ the area of the orthogonal projection of K onto the plane perpendicular to $\mathbf{u}$.

$$\frac{1}{4\pi} \int_{S^2} A(\mathbf{u}) d\Omega(\mathbf{u}) = \frac{S}{4}.$$

For a fixed direction $\mathbf{u}$, lines parallel to $\mathbf{u}$ intersect K in chords of length $l_{\mathbf{u}}(x)$, where x denotes the projection point. Since volume equals the integral of chord lengths over the projection area,

$$\int_{\text{projection}} l_{\mathbf{u}}(x) dA(x) = V,$$

the mean chord length in this direction is

$$L(\mathbf{u}) = \frac{V}{A(\mathbf{u})}.$$

Choosing $\mathbf{u}$ uniformly on the unit sphere, the overall mean chord length is

$$\langle l \rangle = \frac{\int_{S^2} V d\Omega(\mathbf{u})}{\int_{S^2} A(\mathbf{u}) d\Omega(\mathbf{u})} = \frac{4\pi V}{\pi S} = \frac{4V}{S}.$$

For a sphere of radius R , we have $V = \frac{4}{3}\pi R^3$ and $S = 4\pi R^2$, hence

$$\langle l \rangle = \frac{4}{3}R.$$

in the mean time y and z satisfy

$$y = \frac{\Delta E}{\langle \ell \rangle} = \frac{\Delta E}{4r/3}, \quad z = \frac{\Delta E}{\rho_w V} = \frac{y}{\pi r^2 \rho_w}$$

which leads to the relationship between $y(\text{keV}/\mu\text{m})$, $r(\mu\text{m})$ and $z(\text{Gy})$

$$z = 0.204 \frac{y}{(2r)^2}$$

7.3.2 Microdosimetric spectra

Measured spectra: $f(y)$ (frequency) and $d(y)$ (dose). Experimental TEPC/Si pulse-height spectra are converted to tissue-equivalent y by application of W -value and density corrections. Typical behaviors:

- Proton spectra shift right and develop a high- y tail toward and beyond the Bragg peak.
- Heavy ions and HZE particles produce pronounced high- y tails (clustered events).
- Electrons at low energies exhibit condensed-phase effects (liquid vs vapor) that lower y_F and y_D in nm-sites.

7.4 Measurement of microdosimetry

7.4.1 Instruments and conversions

TEPC (Tissue-Equivalent Proportional Counter).

- Uses low-pressure tissue-equivalent gas to simulate a chosen site size.
- $W_{\text{gas}} \sim 26 \text{ eV}$ per ion pair (used to convert pulse-height to energy).
- Challenges: wall effects, δ -ray escape, calibration to tissue.

Silicon microdosimeters (SOI, micro-trench).

- Define precise sensitive volumes (example: $20 \times 20 \times 5 \mu\text{m}^3$).
- $W_{\text{Si}} \approx 3.65 \text{ eV/pair}$; conversion to tissue requires w -value correction and response-function calibration.
- Typically underestimate low- y contribution (insensitive to long-range δ -rays).

Nanodosimeters.

- Count ionisation cluster sizes n in nm-scale volumes (GEM, track-etched, single-ion counting).
- Crucial for Auger electron and nm-scale DNA damage assessment.

7.5 Biological effect and radiation quality

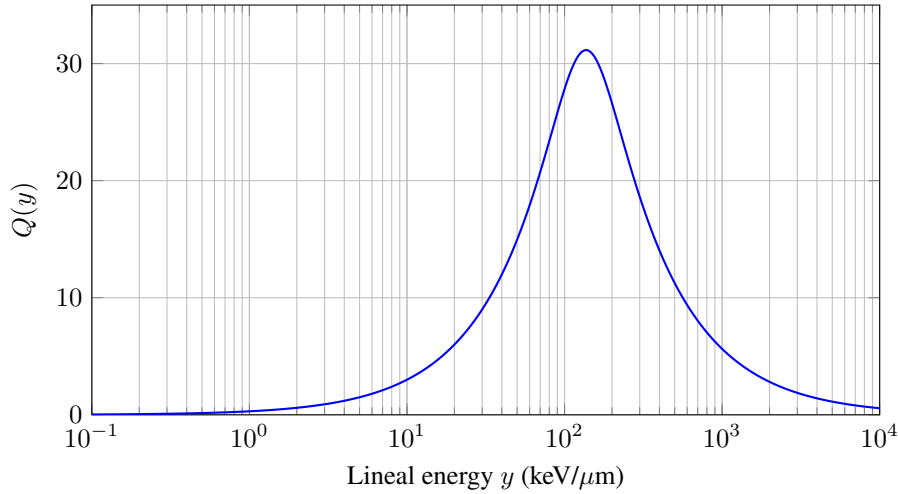
Radiation quality metrics derived from microdosimetry are better correlated with RBE than LET alone:

- Low-LET regime (electrons, $\bar{y}_D < 10 \text{ keV}/\mu\text{m}$): sparse tracks, RBE 1.
- Intermediate (protons): platform $\bar{y}_D \sim 2\text{--}5 \text{ keV}/\mu\text{m}$; rises to tens of $\text{keV}/\mu\text{m}$ at Bragg peak distal edge; proton RBE commonly 1.0–1.2, can reach 1.2–1.5 distally.
- High-LET (heavy ions): $\bar{y}_D > 100 \text{ keV}/\mu\text{m}$; clustered damage, RBE increases then falls under overkill.

ICRP also provides functional forms of $Q(\text{LET})$ for radiation protection purposes. A commonly used ICRP relation is

$$Q(y) = 0.3 y \left[1 + \left(\frac{y}{137} \right)^5 \right]^{-0.4},$$

with y expressed in $\text{keV}/\mu\text{m}$.



The mean relative biological effectiveness can be defined as

$$\bar{r} = \int_0^\infty r(y) d(y) dy,$$

where $r(y)$ is the effectiveness function and $d(y)$ the dose distribution in lineal energy. The RBE is then $y d(y) \cdot r(y)$.

Finally, the equivalent dose H is defined as

$$H = \bar{Q} D,$$

where $\bar{Q} = \int_0^\infty Q(y) d(y) dy$ is the dose-averaged quality factor and D the absorbed dose.

7.6 Microdosimetric Kinetic Model (MK)

MK links stochastic energy deposition to lesion kinetics in domains in $S = \exp(-L)$, $L \propto Az + Bz^2$ where A is from MKM and B is from TDRA. Key ingredients:

$$\begin{aligned} \eta_L(z) &= \eta_0 (1 - e^{-\beta_0 z}), \\ \eta_{QL}(z) &= 2\eta_0 (e^{-\beta_0 z} - e^{-2\beta_0 z}), \end{aligned}$$

where η_0 is initial lesion yield per Gy per domain and β_0 parametrises saturation. Representative mapping to LQ parameters (schematic):

$$\alpha = 2\eta_0\epsilon\left(1 + \frac{\kappa z_1}{2}\right), \quad \beta = \eta_0^2\epsilon^2\kappa,$$

with $\epsilon = \frac{z_1}{2D_0}$, κ an interaction probability and z_1 the single-event mean specific energy. MK thus predicts α, β as functions of microdosimetric spectra (via z_1 , moments, and interaction parameters) and explains dose-dependent RBE and saturation/overkill effects.

The linear coefficient α may be written as a sum of a baseline term and a contribution proportional to a saturation-corrected lineal-energy measure y^* :

$$\alpha = \alpha_0 + \frac{\beta}{\rho \pi r_d^2} y^*,$$

where

- α_0 is the baseline linear damage coefficient (e.g. for reference radiation or limiting low- y behavior);
- β is the quadratic LQ coefficient (pairwise/interaction term);
- ρ is the medium density (e.g. tissue density, with SI units kg m^{-3} or g cm^{-3} depending on convention);
- r_d is an effective domain radius (length scale of the biological domain, e.g. a subcellular or DNA-related domain);
- y^* is a saturation-corrected lineal-energy descriptor ($\text{keV}/\mu\text{m}$).

A commonly used saturation-corrected moment y^* is defined by

$$y^* = \frac{y_0^2 \int_0^\infty \left[1 - \exp\left(-\left(\frac{y}{y_0}\right)^2\right)\right] f(y) dy}{\int_0^\infty y f(y) dy},$$

where

- $f(y)$ is the frequency distribution of lineal energy events;
- y_0 is a saturation parameter (characteristic lineal energy) that controls the transition from linear to saturated response.

Remarks and limiting behaviour

- For small lineal energies $y \ll y_0$, the integrand behaves as $1 - \exp\left(-\left(y/y_0\right)^2\right) \approx \left(y/y_0\right)^2$, so y^* is small and $\alpha \approx \alpha_0$ (no saturation).
- For large $y \gg y_0$, the bracket tends to unity and the numerator approaches $y_0^2 \int f(y) dy = y_0^2$, hence y^* approaches $y_0^2/\bar{y}_F$ (where $\bar{y}_F = \int y f(y) dy$), reflecting saturated (overkill-like) weighting of high- y events.
- The prefactor $\beta/(\rho \pi r_d^2)$ provides dimensional consistency and encodes the coupling between microdosimetric energy density (via y^*) and the macroscopic LQ parameter α . Choosing the appropriate units for ρ and r_d is important to ensure α has units of inverse dose (Gy^{-1}).

7.7 Useful numerical examples and orders of magnitude

(Conversions: $1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$; water density $\rho \approx 1000 \text{ kg m}^{-3}$.)

1 μm diameter sphere (radius $R = 0.5 \mu\text{m}$).

$$V = \frac{4}{3}\pi R^3 \approx 5.236 \times 10^{-19} \text{ m}^3, \quad m = \rho V \approx 5.236 \times 10^{-16} \text{ kg}.$$

A single 1 keV deposition ($\epsilon = 1 \times 10^3 \text{ eV} = 1.602 \times 10^{-16} \text{ J}$) gives

$$z = \frac{\epsilon}{m} \approx 0.306 \text{ Gy}.$$

2 nm diameter sphere (radius $R = 1 \text{ nm}$).

$$V \approx 4.189 \times 10^{-27} \text{ m}^3, \quad m \approx 4.189 \times 10^{-24} \text{ kg},$$

so a 1 keV deposition yields

$$z \approx 3.82 \times 10^7 \text{ Gy}.$$

Remark: At nanometer scales, expressing single-event local energy deposition in units of Gy yields very large numerical values; this should be interpreted as an instantaneous, highly localized energy density rather than a macroscopic averaged dose.

7.8 Typical ranges and scale

Quantity	Typical range / remark
Site size	nm: 1–10 nm (DNA); μm : 0.1–10 μm (nucleus, organelles)
y_D (low-LET)	< 10 keV/ μm
Proton platform y_D	$\sim 2\text{--}5$ keV/ μm
Proton distal y_D	$\sim 20\text{--}100$ keV/ μm (Bragg peak distal edge)
High-LET heavy ions	> 100 keV/ μm (clustered damage)
W -values	gas: ~ 26 eV/pair; Si: ~ 3.65 eV/pair
Proton RBE	$\sim 1.0\text{--}1.2$ (typical), up to ~ 1.5 distally (cell-type dependent)

7.9 Practical modelling / measurement checklist

1. Define site geometry precisely and compute mean chord length (sphere: $\langle l \rangle = 4R/3$).
2. For TEPC measurements, apply wall-effect and δ -ray escape corrections; for Si detectors, apply w -value and tissue-equivalence calibration.
3. For nanometer-scale problems, use condensed-phase cross sections or apply experimental corrections (gas $\rightarrow$ liquid conversion).
4. Report both frequency spectrum $f(y)$ and dose spectrum $d(y)$, and use spectral moments ($\bar{y}_F, \bar{y}_D$) as model inputs (e.g., for MK model).
5. In SOBP or mixed-field situations, perform spectral decomposition to assess distal/high- y components and their impact on RBE.

8 Neutron Interaction with Matter

Neutrons are uncharged subatomic particles with a mass approximately equal to that of the proton. Unlike charged particles, neutrons interact with matter primarily via nuclear forces rather than electromagnetic interactions. As a result, they cause indirect ionization through secondary charged particles, which deposit energy and induce biological damage in tissues. The following subsections describe the mechanisms, classifications, and dosimetric quantities relevant to neutron interactions.

8.1 Local Effect Model: Cell Damage by Ions

The *Local Effect Model* (LEM) is a biophysical framework developed to predict the relative biological effectiveness (RBE) of ionizing radiation, particularly for high-LET (linear energy transfer) particles such as those produced by neutron interactions. Originally developed for carbon ion therapy, LEM is also applicable to neutron-induced recoil ions. The model assumes that cellular damage, especially DNA double-strand breaks (DSBs), depends on the local dose distribution within the cell nucleus rather than the macroscopic average dose.

Neutron interactions produce recoil ions (e.g., protons from elastic scattering with hydrogen) that generate dense ionization tracks, enhancing clustered damage. Key assumptions of LEM include:

- The cell nucleus is modeled as a cylinder or sphere subdivided into nanometer-scale voxels.
- The local dose in each voxel determines the number of lethal lesions, referenced to a photon dose-response curve.
- For ions, the radial dose profile around the track core is crucial, leading to overkill effects at high LET.

For uniform dose D , the cell survival fraction $S(D)$ follows the linear-quadratic (LQ) model:

$$S(D) = \exp(-\alpha D - \beta D^2),$$

where α and β are derived from photon irradiation data (e.g., $\alpha = 0.1 \text{ Gy}^{-1}$, $\beta = 0.05 \text{ Gy}^{-2}$ for typical cell lines).

The local dose $d(r)$ at a radial distance r from a particle track can be approximated as:

$$d(r) = \frac{\text{LET}}{2\pi\rho r^2}, \quad r > r_{\min},$$

where ρ is the medium density ($\sim 1 \text{ g/cm}^3$ for water), LET is in keV/ μm , and $r_{\min} \sim 10 \text{ nm}$ is the track core radius.

The average number of lethal events per nucleus can then be expressed as:

$$N = \int_V [\alpha d(r) + \beta d(r)^2] dV, \quad \text{with survival fraction } S = e^{-N}.$$

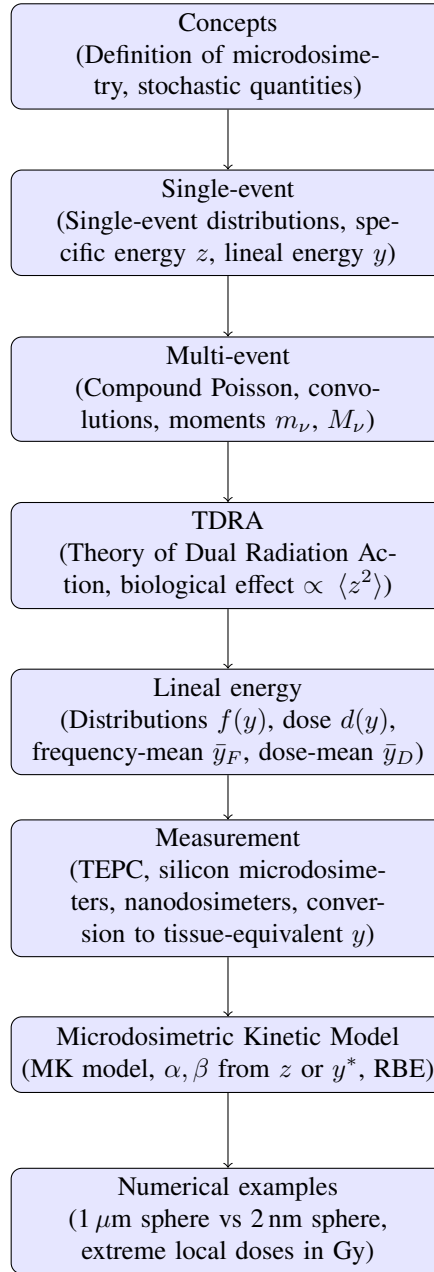


Figure 2: Microdosimetry logical flow: from concepts to numerical examples.

For a more general, three-dimensional spatial dose distribution $D(\vec{r})$, the lethal events can be written as:

$$N = \frac{\iiint_V [\alpha D(\vec{r}) + \beta D(\vec{r})^2] d^3r}{V}.$$

When considering subnuclear domains (e.g., chromatin or molecular volumes), the microscopic average number of lethal events and survival fraction are:

$$\begin{aligned}\bar{N}_C &= -\frac{1}{V_{\text{mol}}} \int_{V_{\text{mol}}} \ln S_x(d_c(\vec{r})) d^3r, \\ \bar{z}_c &= \frac{1}{V_{\text{mol}}} \int_{V_{\text{mol}}} d_c(\vec{r}) d^3r, \quad S_c = \exp(-\bar{N}_C),\end{aligned}$$

where V_{mol} is the subnuclear (molecular) volume, $d_c(\vec{r})$ is the local dose in this domain, and $S_x(d_c)$ is the reference photon survival for dose d_c .

For neutrons, recoil protons typically have LET values of 20–100 keV/ μm , resulting in RBE ~ 1.5 –3 and reducing the oxygen enhancement ratio (OER) to ~ 1.5 , compared to 3 for photons, which is advantageous for hypoxic tumors. Modern LEM versions incorporate refinements to address overestimation at high LET; LEM IV explicitly considers cluster effects for improved accuracy in neutron therapy planning.

Table 4: LEM Parameters for Neutron-Induced Recoils vs. Photons

Parameter	Photons	Proton Recoils (10–50 keV/ μm)	Alpha Particles (>100 keV/ μm)
α (Gy^{-1})	0.1–0.3	0.2–0.5	0.4–0.8
β (Gy^{-2})	0.01–0.05	0.005–0.02	Negligible (overkill)
RBE (at 2 Gy)	1	1.5–2.5	3–5

8.2 Classification of Neutron Interactions

Neutron interactions depend strongly on their kinetic energy:

- **Thermal neutrons:** $E \leq 0.5$ eV (typically 0.025 eV at 20°C), high capture probability.
- **Intermediate neutrons:** 0.5 eV – 10 keV, resonance capture dominant.
- **Fast neutrons:** > 10 keV, scattering dominant, used in therapy for deep penetration.

Neutron interactions produce charged secondary particles for detection and dosimetry.

8.2.1 Elastic Scattering

Elastic scattering conserves kinetic energy and momentum:

$$n + A \rightarrow n' + A',$$

with energy transferred to the recoil nucleus given by

$$E_t = E_n \frac{4M_n M_A}{(M_n + M_A)^2} \cos^2 \theta,$$

where

- E_n is the incident neutron energy,
- M_A is the target nucleus mass in atomic mass units (u),
- θ is the recoil angle in the laboratory frame.

Maximum energy transfer occurs for hydrogen ($M_A = 1$ u), with

$$E_t^{\text{max}} = E_n.$$

In tissues, this process produces recoil protons with ranges up to centimeters for MeV neutrons. Typical elastic scattering cross-sections for light nuclei are $\sigma_{el} \sim 1$ –10 b.

The average energy transferred per collision can be approximated by

$$\overline{E}_t = E_n \frac{2M_n M_A}{(M_n + M_A)^2}.$$

The corresponding KERMA (kinetic energy released per unit mass) is

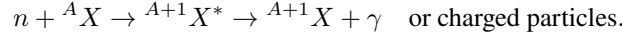
$$K(E_n) = 1.60 \times 10^{-6} E_n \sum_i N_i \epsilon_i \sigma_i,$$

where

$$\epsilon_i = \frac{E_t}{E_n}, \quad N_i \text{ is the number of target nuclei per unit mass, and } \sigma_i \text{ the cross-section for reaction } i.$$

8.2.2 Neutron Capture

Neutron capture reactions generally follow:



For thermal neutrons, the cross-section approximately follows the $1/v$ law:

$$\sigma_c \propto \frac{1}{\sqrt{E_n}}.$$

Typical capture reactions in tissue include:

- Hydrogen: ${}^1\text{H}(n, \gamma){}^2\text{H}$, $Q = 2.225 \text{ MeV}$, $\sigma = 0.33 \text{ b}$.
- Nitrogen: ${}^{14}\text{N}(n, p){}^{14}\text{C}$, $Q = 0.626 \text{ MeV}$, $\sigma = 1.81 \text{ b}$, proton energy $E_p = 0.58 \text{ MeV}$, proton range $\sim 10 \mu\text{m}$, LET $\approx 50 \text{ keV}/\mu\text{m}$.

Because proton ranges are very short, charged particle equilibrium (CPE) can be assumed. The absorbed dose rate D from neutron capture can be estimated as:

$$D = \frac{N \sigma \phi_n E_r \cdot 1.602 \times 10^{-13}}{\rho},$$

where

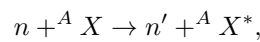
- N is the target atom number density (atoms/cm³),
- σ is the neutron capture cross-section (cm²),
- ϕ_n is the neutron fluence (n/cm²),
- E_r is the energy released per reaction (MeV),
- ρ is the mass density of the medium (g/cm³).

Example: ${}^{14}\text{N}(n, p){}^{14}\text{C}$ in striated muscle:

- $N = 1.504 \times 10^{21} \text{ cm}^{-3}$,
- $\sigma = 1.827 \text{ b} = 1.827 \times 10^{-24} \text{ cm}^2$,
- $E_r = 0.626 \text{ MeV}$,
- $\phi_n = 10^{12} \text{ cm}^{-2}$,
- $\rho = 1 \text{ g/cm}^3$.

8.2.3 Inelastic Scattering

A neutron may undergo **transitory capture** and **reemission**. Scattered neutrons could be in **above ground state**, and as a result, γ -emitting of high energy, that is the main result in **dose deposition**. The energy of the neutron should be **greater than threshold**. Timescale for C, N, O is complicated.



with X^* de-exciting via γ emission. Threshold energy corresponds to the first excited state (e.g., 4.44 MeV for ${}^{12}\text{C}$). Cross-sections $\sigma_{inel} \sim 0.1\text{--}1 \text{ b}$ for light nuclei above threshold.

Example:

$$\begin{aligned} {}^{12}\text{C}(n, n') {}^{12}\text{C}^* &\Rightarrow E_\gamma = 4.43 \text{ MeV} \\ {}^{14}\text{N}(n, n') {}^{14}\text{N}^* &\Rightarrow E_\gamma = 1.63 \text{ MeV}, E_r = 10.0 \text{ MeV}, E_f = 14 \text{ MeV} \\ {}^{16}\text{O}(n, n') {}^{16}\text{O}^* &\Rightarrow E_\gamma = 6.1 \text{ MeV} \\ &\text{etc.} \end{aligned}$$

8.2.4 Nonelastic Scattering

Important for $E_n > 5$ MeV, contribution will be increased as energy increased up to 15 MeV. Nonelastic reactions occur when the kinetic energy of the incident neutron is not fully conserved in the outgoing particles. These processes include capture, inelastic scattering, and reactions like (n, p) , (n, α) , and $(n, 2n)$.

- Important feature: Nonelastic reactions produce not only gamma rays but also high-LET charged particles (protons, alphas), contributing significantly to local dose.
- Example: $^{12}\text{C}(n, n')3\alpha$ for $E_n \gtrsim 10$ MeV, producing high RBE radiation.

Energy partition in a two-body reaction:

$$E_3 + E_2 = Q + E_1,$$

$$E_2 = \frac{2M_1M_2E_1}{(M_2 + M_3)^2} + \frac{M_3Q + E_1(M_3 - M_1)}{M_2 + M_3},$$

where E_1 is incident neutron energy, E_2 and E_3 are kinetic energies of reaction products, μ_i are masses, and Q is the reaction Q-value.

8.2.5 Spallation

Spallation is a process in which a high-energy neutron fragments a heavy nucleus, ejecting multiple nucleons and light charged particles. Key points:

- Significant only for neutron energies $\gtrsim 100$ MeV.
- Produces many charged fragments, leading to high local energy deposition.
- The cross-section increases from about 100 MeV to several hundred MeV.
- These reactions are important in accelerator-based neutron fields for dose estimation, especially for high-LET contributions.

8.3 Neutron KERMA and Dose

8.3.1 KERMA

KERMA (Kinetic Energy Released per unit MAss) quantifies the initial kinetic energy transferred to charged particles per unit mass:

$$K = \Phi \sum_i f_i \bar{E}_{\text{tr},i},$$

where

- Φ is the neutron fluence (n/cm^2),
- f_i is the mass attenuation coefficient for interaction i (cm^2/g),
- $\bar{E}_{\text{tr},i}$ is the average energy transferred to charged particles in interaction i (MeV).

For absorbed dose in cGy, considering interactions of type- j neutrons with type- i target atoms:

$$K^{(ij)} = 1.602 \times 10^{-8} N_i \int_0^{E_{\max}} \frac{d\phi}{dE} \sigma^{(ij)}(E) E_t^{(ij)}(E) dE,$$

where

- N_i is the number density of type- i atoms (atoms/cm^3),
- $\frac{d\phi}{dE}$ is the neutron differential fluence ($\text{n/cm}^2/\text{MeV}$),
- $\sigma^{(ij)}(E)$ is the microscopic cross-section for the i - j interaction (cm^2),
- $E_t^{(ij)}(E)$ is the average energy transferred per interaction (MeV),
- $E_{\max}$ is the maximum neutron energy considered.

Table 5: Neutron KERMA Factors (10^{-12} Gy m²) for Tissue Elements

E_n (MeV)	H	C	N	O
Thermal (0.025 eV)	0.00033	0	2.74	0
1	12	5.4	4.8	4.2
10	25	11	9.6	8.5

For thermal neutrons in muscle (composition H:C:N:O = 0.102:0.123:0.035:0.74):

- From ¹⁴N: $k_N = 2.74 \times 10^{-11}$ rad cm²/n.
- From ¹H: $k_H = 3.32 \times 10^{-15}$ Gy/atom (γ only).

Total KERMA $K \sim 1.6 \times 10^{-6}$ Gy m²/n. Table 5 lists k_f for tissue elements.

8.3.2 Photon Dose from Secondary Gamma

For secondary gamma rays produced in neutron interactions, the dose can be expressed using mass energy-absorption coefficients:

$$D_\gamma = \Phi(E_\gamma) E_\gamma \left(\frac{\mu_{\text{en}}}{\rho} \right)_{E_\gamma},$$

where

- $\Phi(E_\gamma)$ is the gamma fluence (photons/cm²),
- E_γ is the photon energy (MeV),
- $(\mu_{\text{en}}/\rho)_{E_\gamma}$ is the mass energy-absorption coefficient of the medium at energy E_γ (cm²/g).

8.3.3 Dose Equivalent and Quality Factor

To account for biological effectiveness, the neutron dose is weighted by a quality factor $w(E)$:

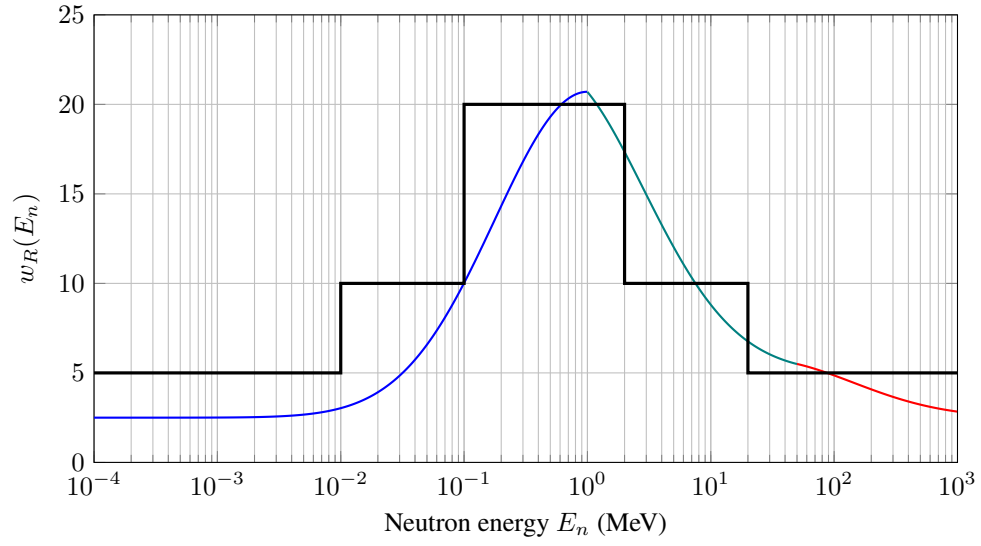
$$H_n = \bar{w} D_n, \quad \bar{w} = \frac{\int_0^{E_{\text{max}}} w(E) \frac{dD}{dE} dE}{\int_0^{E_{\text{max}}} \frac{dD}{dE} dE}.$$

where

- H_n is the neutron dose equivalent (Sv),
- D_n is the neutron absorbed dose (Gy or cGy),
- $\bar{w}$ is the fluence- or dose-weighted average quality factor.

ICRP recommendations (ICRP 60/103):

$$w_R(E) = \begin{cases} 2.5 + 18.2 \exp\left[-\frac{(\ln E)^2}{6}\right], & E < 1 \text{ MeV}, \\ 5.0 + 17.0 \exp\left[-\frac{(\ln(2E))^2}{6}\right], & 1 \leq E \leq 50 \text{ MeV}, \\ 2.5 + 3.25 \exp\left[-\frac{(\ln(0.04E))^2}{6}\right], & E > 50 \text{ MeV}. \end{cases}$$



— ICRP: $E < 1$ MeV — ICRP: $1 \leq E \leq 50$ MeV
 — ICRP: $E > 50$ MeV — Discrete step values (table)

Neutron Energy E_n	Quality Factor w_R
< 10 keV	5
10–100 keV	10
0.1–2 MeV	20
2–20 MeV	10
> 20 MeV	5

Table 6: Discrete values of neutron quality factor.

9 Neutron Dosimetry

9.1 Equation for n - γ Response

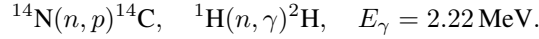
Neutron dosimetry aims to quantify absorbed dose D in mixed neutron-photon fields, often separating contributions based on relative biological effectiveness (RBE):

$$R = \alpha D_\gamma + \beta D_n,$$

where α and β weight the photon and neutron dose components, respectively.

9.2 Thermal Neutron Interactions

Thermal neutrons interact with matter primarily through capture and nuclear reactions. Two important examples in tissue are:



The $^{14}\text{N}(n, p)^{14}\text{C}$ reaction converts nitrogen into carbon while emitting a proton. The $^1\text{H}(n, \gamma)^2\text{H}$ reaction is a neutron capture process where hydrogen absorbs a neutron and releases a photon of energy 2.22 MeV.

These reactions are biologically important because they transfer energy locally, contributing to dose.

9.2.1 Kerma from $^{14}\text{N}(n, p)^{14}\text{C}$

The KERMA (Kinetic Energy Released per unit Mass) quantifies the energy transferred from neutrons to charged particles per unit mass:

$$K = \Phi_n F_n,$$

where Φ_n is the neutron fluence and F_n is the kerma factor.

For nitrogen capture:

$$F_n = 2.74 \times 10^{-11} \text{ rad}, \quad \sigma_N = 1.8 \times 10^{-24} \text{ cm}^2, \quad N_\mu = 4.5 \times 10^{21} \text{ atoms/g}.$$

These numbers reflect the low probability of neutron capture by nitrogen compared to hydrogen.

9.2.2 Energy Transfer from $^1\text{H}(n, \gamma)^2\text{H}$

When hydrogen captures a neutron, the emitted photon carries away:

$$E_{\text{tr}} = E_\gamma = 2.22 \text{ MeV per neutron}.$$

Although nitrogen also reacts, hydrogen is far more abundant in tissue and therefore dominates the energy transfer.

9.2.3 Tissue Composition and Atom Density

In muscle tissue (per gram of material), the composition is approximately:

$$\text{H: } 0.102 \text{ g}, \quad \text{C: } 0.123 \text{ g}, \quad \text{N: } 0.035 \text{ g}, \quad \text{O: } 0.74 \text{ g}.$$

The number of atoms of an element i per gram is calculated as:

$$N_i = \frac{m_i}{A_i} N_A,$$

with $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ the Avogadro constant.

Because hydrogen atoms are very numerous relative to nitrogen, the total KERMA from $^1\text{H}(n, \gamma)$ is about 26 times larger than from $^{14}\text{N}(n, p)$. This is true for any neutron field that eventually thermalizes.

9.3 Neutron Quality Factor

The equivalent dose H is used to weight the absorbed dose K by a quality factor Q that reflects biological effectiveness:

$$H = QK,$$

where H is measured in sieverts.

For photons, $Q_\gamma \approx 1$, but for neutrons Q varies with energy because neutron interactions produce higher linear energy transfer (LET) particles.

At very low neutron energies ($E_n \leq 30 \text{ eV}$), the main contribution comes from the $^{14}\text{N}(n, p)^{14}\text{C}$ reaction. At higher energies ($E_n > 30 \text{ eV}$), elastic scattering with hydrogen, carbon, nitrogen, and oxygen dominates energy transfer.

9.4 Dosimetry in Mixed Neutron–Photon Fields

Even a source that emits only neutrons will inevitably produce photons when neutrons interact with matter. This means radiation fields are almost always mixed ($n + \gamma$), and dosimeters must respond to both components.

The general response of a dosimeter in a mixed field is:

$$R_{n,\gamma} = AD_\gamma + BD_n,$$

where D_γ and D_n are the absorbed doses from photons and neutrons, and A, B are sensitivity coefficients.

For tissue-equivalent (TE) ion chambers, a normalized form is used:

$$R'_{n,\gamma} = hD_\gamma + kD_n,$$

with

$$h = \frac{A}{S_\gamma}, \quad k = \frac{B}{S_\gamma}.$$

Because TE chambers are made of low- Z materials, $h \approx 1$, meaning their gamma response is nearly ideal.

9.5 Paired Dosimeters

To separate photon and neutron dose components, two detectors with different sensitivities can be used together. For example:

- Dosimeter 1: Tissue-equivalent ionization chamber, sensitive to both photons and neutrons.
- Dosimeter 2: Thermoluminescent dosimeter (TLD) without hydrogen, mainly sensitive to photons.

Their responses can be written as:

$$R_{n,\gamma}^{\text{TE}} = D_\gamma + \left(\frac{B}{A}\right)_{\text{TE}} D_n,$$

$$R_{n,\gamma}^{\text{TLD}} = D_\gamma + \left(\frac{B}{A}\right)_{\text{TLD}} D_n.$$

By combining the two equations, D_γ and D_n can be separated and solved for.

9.6 Calibration of a Tissue-Equivalent Ion chamber for $n + \gamma$ dosimetry

9.6.1 Exposure in Air from a ^{60}Co Source

The absorbed dose to tissue from a ^{60}Co gamma source, expressed using the Tissue-Equivalent Calibration Protocol (TCPE), is given by:

$$D_\gamma \stackrel{\text{TCPE}}{=} \beta A_{\text{eq}} X \left(\frac{\overline{W}}{e}\right)_{\text{air}} \left(\frac{\mu_{\text{en}}}{\rho}\right)_{\text{air}}^{\text{tiss}},$$

where

- $\beta = 1.003$ is the calibration correction factor,
- X is the exposure in air [C/kg],
- $\left(\frac{\overline{W}}{e}\right)_{\text{air}} = 33.97 \text{ J/C}$ is the mean energy expended per unit charge of ionization in air,
- $\left(\frac{\mu_{\text{en}}}{\rho}\right)_{\text{air}}^{\text{tiss}} = \frac{0.0293}{0.0266} = 1.102$ is the ratio of mass energy-absorption coefficients (tissue-to-air), taken from Attix for ^{60}Co photons.

Substituting these values yields:

$$D_\gamma = 37.1 X \quad [\text{Gy}],$$

9.6.2 TE ion chamber equivalent

9.6.3 Tissue-Equivalent Ion Chamber Response

For a tissue-equivalent (TE) ionization chamber exposed to photons, the measured charge is denoted as

$$(Q_\gamma)_{\text{TE}}.$$

The photon sensitivity of the chamber, A_{TE} , is defined as the ratio of collected charge to absorbed dose in tissue:

$$A_{\text{TE}} \equiv \frac{(Q_\gamma)_{\text{TE}}}{D_\gamma}.$$

Introducing the calibration relation through the absorbed dose in the chamber medium $(D_\gamma)_{\text{TE}}$, we can rewrite:

$$A_{\text{TE}} = \frac{(Q_\gamma)_{\text{TE}}}{(D_\gamma)_{\text{TE}}} \frac{(D_\gamma)_{\text{TE}}}{D_\gamma}.$$

Applying the TCPE formalism, the ratio of absorbed doses converts via the mass energy-absorption coefficient ratio:

$$A_{\text{TE}} \stackrel{\text{TCPE}}{=} \frac{(Q_\gamma)_{\text{TE}}}{(D_\gamma)_{\text{TE}}} \left(\frac{\mu_{\text{en}}}{\rho} \right)_{\text{tiss}}^{\text{TE}}.$$

9.6.4 Bragg–Gray Cavity Theory

According to Bragg–Gray cavity theory, the absorbed dose in the cavity gas of the TE chamber is

$$D_{g(\text{TE})} = \frac{(Q_\gamma)_{\text{TE}}}{\rho_{g(\text{TE})} V} \left(\frac{\overline{W}_\gamma}{e} \right)_{g(\text{TE})},$$

where $(Q_\gamma)_{\text{TE}}$ is the collected charge, $\rho_{g(\text{TE})}$ the gas density, V the cavity volume, and $\left(\frac{\overline{W}_\gamma}{e} \right)_{g(\text{TE})}$ the mean energy per ion pair in the chamber gas.

The absorbed dose in the chamber wall material (i.e., tissue-equivalent medium) follows from the stopping power ratio:

$$D_{w(\text{TE})} = D_{g(\text{TE})} S_g^{\text{TE}},$$

with S_g^{TE} the mass stopping power ratio (wall-to-gas).

Thus, the charge-to-dose ratio can be written as

$$\frac{(Q_\gamma)_{\text{TE}}}{(D_\gamma)_{\text{TE}}} = \frac{\rho V}{\left(\frac{\overline{W}_\gamma}{e} \right)_g \left(\frac{\overline{S}_\gamma}{\rho} \right)_{g(\text{TE})}^{\text{TE}}}.$$

With the parameters:

$$\frac{\overline{W}_\gamma}{e} = 29.3 \text{ J/C}, \quad \rho = 1.046 \text{ kg/m}^3 \quad (25^\circ\text{C}),$$

the photon sensitivity of the TE ion chamber becomes

$$A_{\text{TE}} = \left(\frac{\mu_{\text{en}}}{\rho} \right)_{\text{tiss}}^{\text{TE}} \frac{\rho V}{\left(\frac{\overline{W}_\gamma}{e} \right)_g \left(\frac{\overline{S}_\gamma}{\rho} \right)_{g(\text{TE})}^{\text{TE}}}.$$

9.6.5 Neutron Response of the TE Chamber

For neutron irradiation, the sensitivity of the TE chamber is defined as

$$B_{\text{TE}} \equiv \frac{(Q_n)_{\text{TE}}}{D_n},$$

where $(Q_n)_{\text{TE}}$ is the collected charge under neutron exposure, and D_n is the absorbed dose to tissue from neutrons.

Introducing the absorbed dose in the chamber medium $(D_n)_{\text{TE}}$, one obtains

$$B_{\text{TE}} = \frac{(Q_n)_{\text{TE}}}{(D_n)_{\text{TE}}} \frac{(D_n)_{\text{TE}}}{D_n}.$$

Within the TCPE formalism, the conversion from chamber medium dose to tissue dose is expressed by the neutron kerma factor ratio:

$$B_{\text{TE}} \stackrel{\text{TCPE}}{=} \frac{(Q_n)_{\text{TE}}}{(D_n)_{\text{TE}}} (F_n)_{\text{tiss}}^{\text{TE}},$$

where $(F_n)_{\text{tiss}}^{\text{TE}}$ accounts for the conversion from absorbed dose in the TE chamber medium to absorbed dose in tissue for neutron fields.

For the present case, the neutron reference dose is taken as

$$D_n = 0.52 \text{ g/cm}^2.$$

9.6.6 Sensitivity

From the Bragg–Gray expressions already obtained for the neutron cases,

$$\frac{(Q_n)_{\text{TE}}}{(D_n)_{\text{TE}}} = \frac{\rho V}{\left(\frac{\overline{W}_n}{e}\right)_g \left(\frac{\overline{S}_n}{\rho}\right)_{g(\text{TE})}^{\text{TE}}}.$$

$$B_{\text{TE}} = \frac{\rho V}{\left(\frac{\overline{W}_n}{e}\right)_g \left(\frac{\overline{S}_n}{\rho}\right)_{g(\text{TE})}^{\text{TE}}} (F_n)_{\text{tiss}}^{\text{TE}}$$

The ratio of neutron-to-photon sensitivities of the TE chamber can be expressed as

$$\frac{B_{\text{TE}}}{A_{\text{TE}}} = (F_n)_{\text{tiss}}^{\text{TE}} \left(\frac{\mu_{\text{en}}}{\rho}\right)_{\text{TE}}^{\text{tiss}} \frac{\left(\frac{\overline{W}_\gamma}{e}\right)_g \left(\frac{\overline{S}_n}{\rho}\right)_{g(\text{TE})}^{\text{TE}}}{\left(\frac{\overline{W}_n}{e}\right)_g \left(\frac{\overline{S}_\gamma}{\rho}\right)_{g(\text{TE})}^{\text{TE}}},$$

Under the usual practical approximations used when quoting a simple conversion factor, the multiplicative correction factors are often close to unity. Specifically, for a TE chamber where the gas and wall are of the same tissue-equivalent material:

$$\left(\frac{\overline{S}_\gamma}{\rho}\right)_{g(\text{TE})}^{\text{TE}} \approx 1, \quad \left(\frac{\overline{S}_n}{\rho}\right)_{g(\text{TE})}^{\text{TE}} \approx 1,$$

since the stopping powers of the TE gas and TE wall are nearly identical. Moreover, in the photon energy range dominated by Compton interactions:

$$\left(\frac{\mu_{\text{en}}}{\rho}\right)_{\text{TE}}^{\text{tiss}} \approx 1.$$

Under the practical approximations discussed above, the neutron-to-photon sensitivity ratio reduces to

$$\frac{B_{\text{TE}}}{A_{\text{TE}}} \approx (F_n)_{\text{tiss}}^{\text{TE}} \frac{(\overline{W}_\gamma/e)_g}{(\overline{W}_\gamma/e)_g} \equiv f(E_n) = \alpha.$$

For neutron energies near the maximum, $E_n^{\text{max}} = 14\text{--}15$ MeV, the mean ionization energy ratio is

$$\frac{(\overline{W}_\gamma/e)_g}{(\overline{W}_\gamma/e)_g} = 1.062,$$

and the neutron dose conversion factor gives $(F_n)_{\text{tiss}}^{\text{TE}} \approx 1.028$.

Thus, the overall sensitivity ratio becomes

$$\frac{B_{\text{TE}}}{A_{\text{TE}}} \approx 1.028 \times 1.062 \approx 0.97.$$

For a TE ionization chamber, this implies

$$B_{\text{TE}} = \alpha A_{\text{TE}} \approx A_{\text{TE}},$$

so that the neutron and photon sensitivities are nearly equal in practice.

9.7 Detectors

9.7.1 A-150 TE Plastic Ion Chamber

The A-150 plastic (C:77.7%, H:10.1%, N:3.5%, O:5.2%, F:1.7%, Ca:1.8% by mass) closely mimics tissue in terms of KERMA response. The absorbed dose is determined from the collected charge Q :

$$D = \frac{Q}{m} \cdot \frac{W}{e},$$

where m is the sensitive volume mass, and $W \approx 33.97$ J/C for air. Calibration is usually performed using ^{60}Co gamma rays, with corrections for tissue equivalence and fluence distribution in the chamber gas.

9.7.2 Rossi TE Proportional Counter

A spherical counter (diameter 2.54 cm) with A-150 wall and propane-TE gas measures the lineal energy y :

$$y = \frac{\epsilon}{\bar{l}}, \quad \bar{l} = \frac{2d}{3},$$

for a sphere of diameter $d \sim 1\text{--}10 \mu\text{m}$. Event spectra allow discrimination between neutrons (high- y events) and photons (low- y events).

Dose estimation from spectra:

$$D = \sum_i N_i E_i, \quad E_i = \frac{h_i}{h_\alpha} E_\alpha,$$

where h_i is the event count in channel i , h_α is a calibration factor, and E_α is the reference energy.

9.7.3 Tissue-Equivalent Plastic Calorimeter

Absolute dose can be measured via temperature rise:

$$D = c \frac{\Delta T}{1 - \delta},$$

with $\delta \sim 4\%$ accounting for heat defect. For neutron beams:

$$\frac{D_{n,\text{muscle}}}{D_{n,\text{A-150}}} = \frac{(F_n)_{\text{muscle}}}{(F_n)_{\text{A-150}}},$$

where F_n is the neutron KERMA factor and Φ_n the neutron fluence.

9.7.4 Thermoluminescent Dosimeters (TLD) and GM Tubes

- TLDs: Sensitive to both neutrons and photons, with β component typically $\sim 1/5$ of α component.
- GM tubes: Primarily gamma-sensitive, used for photon monitoring.

9.7.5 Metal Foil Activation Detectors

For neutron activation measurements:

$$C(\tau, t) = \epsilon A'(T, t) = \epsilon N'_t (1 - e^{-\lambda t}) e^{-\lambda \tau} \int_0^\infty \sigma(E) \frac{d\varphi}{dE} dE,$$

where

- $C(\tau, t)$ is the measured count rate at time τ after irradiation of duration t ,
- ϵ is the detector efficiency,
- N'_t is the number of target nuclei,
- λ is the decay constant,
- $\sigma(E)$ is the activation cross-section,
- $d\varphi/dE$ is the neutron energy differential fluence.

9.8 Neutron Spectrum Acquisition Methods

The absorbed dose in tissue can be expressed as:

$$D_{\text{tissue}} = \int \Phi(E) K(E) dE,$$

where $\Phi(E)$ is the neutron energy spectrum (fluence) and $K(E)$ is the corresponding KERMA factor. The challenge is to determine $\Phi(E)$ experimentally.

9.8.1 Time-of-Flight (TOF) Method

Neutron energy is derived from the flight time t over a known path length L :

$$E_n = \frac{1}{2} m_n \left(\frac{L}{t} \right)^2,$$

with energy resolution given by:

$$\frac{\Delta E}{E} = 2 \frac{\Delta t}{t} + 2 \frac{\Delta L}{L}.$$

9.8.2 Bonner Sphere Unfolding Method

Polyethylene spheres (2–18") moderate neutrons to thermal energies, and a central ^6LiI or ^3He detector counts the thermalized neutrons. In discrete form, let N_i be the neutron fluence in energy bin i and n_j the counts recorded by detector j .

The detector response is described by the response matrix α :

$$\begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1M} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{N1} & \alpha_{N2} & \cdots & \alpha_{NM} \end{bmatrix} \begin{bmatrix} N_1 \\ N_2 \\ \vdots \\ N_M \end{bmatrix} = \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_N \end{bmatrix},$$

or component-wise:

$$n_j = \sum_{i=1}^M \alpha_{ji} N_i, \quad j = 1, 2, \dots, N.$$

Here:

- N_i is the neutron fluence in energy bin i (n/cm^2),
- n_j is the measured count in detector j ,
- α_{ji} is the response of detector j to neutrons in energy bin i .

The system is typically solved by least-squares, GRAVEL, or other iterative unfolding methods to obtain the neutron spectrum $\vec{N}$ from the measured counts $\vec{n}$.

9.8.3 Tracking Detectors

Track recoil particles (e.g., in scintillating fibers). For n-p scattering, the neutron energy is obtained from the recoil proton energy E_p and scattering angle ϕ :

$$E_n = \frac{E_p}{\cos^2 \phi}.$$

Two common types of tracking detectors are:

- **Scintillating Fiber Trackers:** Recoil protons produce scintillation light in fine fiber arrays. Systems like SONTRAC reconstruct 3D tracks of particles, providing high spatial resolution and energy determination for neutrons in the 10–100 MeV range.
- **CR-39 Nuclear Track Detectors:** CR-39 is a solid-state polymer that records tracks of charged particles via localized damage. After chemical etching, the tracks become visible under a microscope. The track length and geometry allow reconstruction of the energy and direction of the recoil particle, enabling neutron spectroscopy in high-LET fields. CR-39 is passive, highly sensitive, and suitable for mixed neutron-gamma fields.

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- 11 Bunker shielding**
- 12 Radiation Safety**
- 13 Other topics**
 - 13.1 Nanotechnology Fundamentals and Real Case**
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 - 13.7 Comparison of IMPT with IMRT**

14 Example Exam

14.1 Q1

14.1.1 (a) Regimens R1 and R2

(i) **Fractional clonogen survival** The fractional clonogen survival $S(D)$ is modeled using the linear-quadratic (LQ) model:

$$S(D) = e^{-(\alpha D + \beta D^2)}$$

where D is the dose per fraction, and $\alpha = 0.35 \text{ Gy}^{-1}$, $\beta = 0.035 \text{ Gy}^{-2}$ ($\alpha/\beta = 10 \text{ Gy}$).

Regimen 1 (R1): Total dose = 54 Gy, 3 fractions/day, 1.5 Gy/fraction, 36 fractions over 12 days.

$$S(d) = e^{-(0.35 \cdot 1.5 + 0.035 \cdot 1.5^2)} = e^{-0.60375} \approx 0.546$$

$$S_{\text{total}} = (0.546)^{36} \approx 3.6 \times 10^{-10}$$

Regimen 2 (R2): Total dose = 80.5 Gy, 2 fractions/day, 1.15 Gy/fraction, 70 fractions over 35 days.

$$S(d) = e^{-(0.35 \cdot 1.15 + 0.035 \cdot 1.15^2)} = e^{-0.4488} \approx 0.639$$

$$S_{\text{total}} = (0.639)^{70} \approx 2.28 \times 10^{-14}$$

(ii) **Tumour and late effect BED** BED formula:

$$\text{BED} = nd \left(1 + \frac{d}{\alpha/\beta} \right)$$

- Tumor: $\alpha/\beta = 10 \text{ Gy}$ - Late effects: $\alpha/\beta = 3 \text{ Gy}$

R1:

$$\text{BED}_{\text{tumor}} = 36 \cdot 1.5 \left(1 + \frac{1.5}{10} \right) = 62.1 \text{ Gy}_{10}$$

$$\text{BED}_{\text{late}} = 36 \cdot 1.5 \left(1 + \frac{1.5}{3} \right) = 81 \text{ Gy}_3$$

R2:

$$\text{BED}_{\text{tumor}} = 70 \cdot 1.15 \left(1 + \frac{1.15}{10} \right) = 89.8 \text{ Gy}_{10}$$

$$\text{BED}_{\text{late}} = 70 \cdot 1.15 \left(1 + \frac{1.15}{3} \right) = 111.3 \text{ Gy}_3$$

Interpretation: R2 provides better tumor control (higher tumor BED) but increases late effect risk. R1 favors normal tissue sparing.

(iii) Practical difficulties

- **R1:** 3 fractions/day $\rightarrow$ 12 days, 18 hours total; weekend treatments increase staff workload and patient inconvenience.
- **R2:** 2 fractions/day $\rightarrow$ 35 days, 35 hours total; prolonged schedule challenges patient compliance, but easier for staff planning.

(iv) Time gap and half-life of repair

- **R1:** Time gap = 4 h, $T_{1/2} = 0.7 \text{ h}$, number of half-lives ≈ 5.7
- **R2:** Time gap = 6 h, number of half-lives ≈ 8.6

14.1.2 (b) Tumour Control Probability (TCP)

(i) **TCP for 50 Gy in 25 fractions** Dose/fraction = 2 Gy, clonogens = 10^8 :

$$S(d) = e^{-(0.35 \cdot 2 + 0.035 \cdot 2^2)} = e^{-0.84} \approx 0.432$$

$$S_{\text{total}} = (0.432)^{25} \approx 7.6 \times 10^{-10}$$

$$\text{TCP} = e^{-10^8 \cdot 7.6 \times 10^{-10}} \approx 0.927$$

(ii) Dose for TCP = 0.8

$$\begin{aligned} 0.8 &= e^{-10^8 \cdot S_{\text{total}}} \Rightarrow S_{\text{total}} \approx 2.23 \times 10^{-9} \\ e^{-25(\alpha d + \beta d^2)} &= 2.23 \times 10^{-9} \Rightarrow 0.035d^2 + 0.35d - 0.7848 = 0 \\ d &\approx 1.886 \text{ Gy}, \quad D_{\text{total}} \approx 47.15 \text{ Gy} \end{aligned}$$

(iii) Four Rs of radiobiology

1. Repair: Sublethal DNA damage repair
2. Redistribution: Cell cycle phase affects radiosensitivity
3. Repopulation: Surviving cells proliferate
4. Reoxygenation: Hypoxic cells become oxygenated

(iv) Fifth R: Radiosensitivity Intrinsic susceptibility of cells/tissues to radiation.

14.1.3 (c) Acute radiation syndromes

(i) Three main syndromes

1. Hematopoietic
2. Gastrointestinal (GI)
3. Cerebrovascular/CNS

(ii) Dose thresholds

- Hematopoietic: 1–8 Sv
- GI: 8–30 Sv
- CNS: >30 Sv

(iii) Survival times

- Hematopoietic: weeks–months (50% survival 4–5 Sv)
- GI: 1–2 weeks without intensive care
- CNS: <48 hours fatal

14.2 Q2

14.2.1 (a) Biological Effective Dose (BED) Calculations

(i) Regimen 1: LDR brachytherapy (75 Gy in 75 hours) Using the approximate BED formula for continuous LDR exposure:

$$\text{BED} = D \left(1 + \frac{2R}{\mu \cdot (\alpha/\beta)} \right)$$

where $D = 75 \text{ Gy}$, $T = 75 \text{ h}$, $R = D/T = 1 \text{ Gy/h}$, $\mu = 0.46 \text{ h}^{-1}$ (repair rate, $T_{1/2} \approx 1.5 \text{ h}$).

Tumor: $\alpha/\beta = 10 \text{ Gy}$

$$\text{BED}_{\text{tumor}} = 75 \left(1 + \frac{2 \cdot 1}{0.46 \cdot 10} \right) = 75 \cdot 1.4348 \approx 107.6 \text{ Gy}_{10}$$

Late effects: $\alpha/\beta = 3 \text{ Gy}$

$$\text{BED}_{\text{late}} = 75 \left(1 + \frac{2 \cdot 1}{0.46 \cdot 3} \right) = 75 \cdot 1.4493 \approx 183.7 \text{ Gy}_3$$

(ii) Regimen 2: HDR brachytherapy (40.8 Gy in 6 fractions) Using the standard BED formula for fractionated HDR:

$$\text{BED} = nd \left(1 + \frac{d}{\alpha/\beta} \right)$$

where $n = 6$, $d = 40.8/6 = 6.8 \text{ Gy}$.

Tumor: $\alpha/\beta = 10 \text{ Gy}$

$$\text{BED}_{\text{tumor}} = 6 \cdot 6.8 \left(1 + \frac{6.8}{10} \right) = 40.8 \cdot 1.68 \approx 68.5 \text{ Gy}_{10}$$

Late effects: $\alpha/\beta = 3 \text{ Gy}$

$$\text{BED}_{\text{late}} = 6 \cdot 6.8 \left(1 + \frac{6.8}{3} \right) = 40.8 \cdot 3.2667 \approx 133.3 \text{ Gy}_3$$

(iii) Comparison

- Tumor control BED: R1 (107.6 Gy₁₀) > R2 (68.5 Gy₁₀) by 39.1 Gy₁₀
- Late effect BED: R1 (183.7 Gy₃) > R2 (133.3 Gy₃) by 50.4 Gy₃

(iv) Mechanism in LDR brachytherapy Primarily utilizes **repair**. The low dose rate allows sublethal damage repair in late-responding tissues, sparing normal tissue while continuously damaging tumor cells. Repopulation is minimal during short continuous exposure.

14.2.2 (b) LINAC Primary Barrier Calculations

Given: 6 MV beam, distance $d = 5.305 \text{ m}$, TVL (concrete) = 0.35 m, use factor $U = 1/4$, occupancy $T = 1$, annual workload $W = 50000 \text{ Gy}$, dose limit $P = 0.3 \text{ mSv/year} = 0.0003 \text{ Sv/year}$.

(i) Concrete barrier thickness Transmission factor:

$$B = \frac{P \cdot d^2}{W \cdot U \cdot T} = \frac{0.0003 \cdot 5.305^2}{50000 \cdot 0.25 \cdot 1} \approx 6.75 \times 10^{-7}$$

Number of TVLs:

$$n = -\log_{10} B \approx 6.17$$

Barrier thickness:

$$t = n \cdot \text{TVL} = 6.17 \cdot 0.35 \approx 2.16 \text{ m}$$

(ii) Steel barrier thickness (TVL = 0.055 m)

$$t = 6.17 \cdot 0.055 \approx 0.34 \text{ m}$$

14.2.3 (c) Dose limits and second cancer

(i) ICRP 60 recommended dose limits for occupational exposure

- 20 mSv/year averaged over 5 years
- No single year exceeding 50 mSv

(ii) ALARA principle "As Low As Reasonably Achievable" – optimize radiation exposure considering:

- Economic factors (cost-benefit of dose reduction)
- Social factors (societal benefit vs. risk)

(iii) Criteria for radiation-induced second cancer Based on Cahan et al.:

1. Tumor arises within previously irradiated field
2. Latency period (typically >5 years)
3. Histologically different from primary tumor
4. Not present before radiation
5. No familial syndrome explaining tumor

14.2.4 (d) LQ Model Proofs

(i) **Survival fraction from single to fractionated dose** For single fraction d :

$$s = e^{-(\alpha d + \beta d^2)}$$

For n fractions of dose d each (total dose $D = nd$), assuming complete repair:

$$s_{\text{total}} = \left(e^{-(\alpha d + \beta d^2)} \right)^n = e^{-n(\alpha d + \beta d^2)} = e^{-(\alpha D + \beta D d)}$$

(ii) **Relating $-\ln(s)$ to BED**

$$E = -\ln(s_{\text{total}}) = n(\alpha d + \beta d^2) = \alpha D + \beta D d$$

$$\text{BED} = \frac{E}{\alpha} = D \left(1 + \frac{d}{\alpha/\beta} \right)$$

(iii) **Relating BED to Standard Effective Dose (SED, EQD2)**

$$\text{SED} = \frac{\text{BED}}{1 + \frac{2}{\alpha/\beta}}$$

This converts BED to an equivalent dose delivered in 2 Gy fractions for comparison in non-proliferative scenarios.

14.3 Q3

14.3.1 (a) Radiobiological Effectiveness (RBE)

(i) **Definition of RBE_α and RBE_{10}**

- RBE_α (also RBE_{max}) is the relative biological effectiveness at the low-dose limit:

$$\text{RBE}_\alpha = \frac{\alpha_{\text{test}}}{\alpha_{\text{reference}}}$$

It represents the maximum RBE for high-LET radiation (e.g., neutrons) compared to a low-LET reference (e.g., X-rays), where single-track lethal events dominate.

- RBE_{10} is the RBE at 10% cell survival:

$$\text{RBE}_{10} = \frac{D_{\text{reference}}(S = 0.1)}{D_{\text{test}}(S = 0.1)}$$

It reflects clinically relevant doses.

Graphical representation: - RBE_α : Plot $\ln(S)$ vs dose. X-ray curve has shallow initial slope (α), neutron curve steeper. Ratio of initial slopes gives RBE_α . - RBE_{10} : Isoeffect plot of RBE vs survival/dose; value at $S=0.1$ gives RBE_{10} .

(ii) **X-ray vs fast neutron survival curves**

- X-ray: Sigmoidal on semi-log plot (shoulder due to β term, then linear).
- Fast neutron: Nearly linear exponential (steeper slope, minimal shoulder).

Explanation: Low-LET X-rays produce sparse DNA damage (allowing repair, quadratic β term), while high-LET neutrons cause dense, clustered DNA damage (mostly DSBs, dominated by α term, less repairable).

(iii) **Radiobiological information from curves**

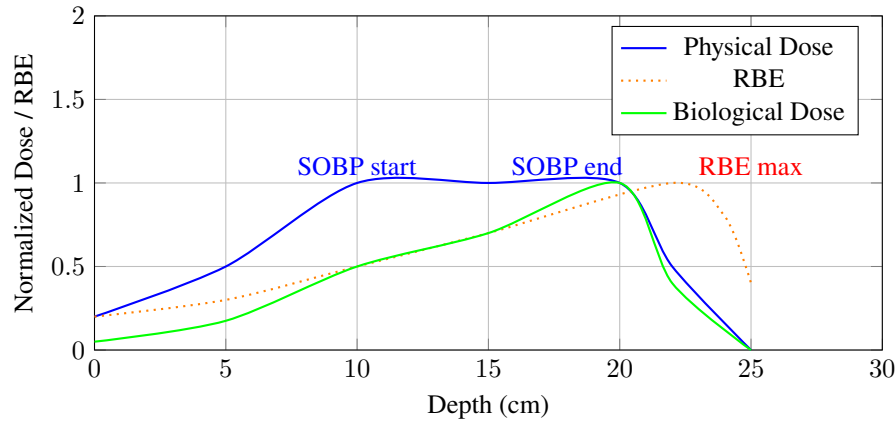
- α : intrinsic radiosensitivity (initial slope)
- β : sublethal damage repair capacity (curvature)
- α/β : tissue fractionation response
- $D_0 = 1/\alpha$: mean lethal dose
- n (extrapolation number): related to repair multiplicity
- OER: oxygen enhancement ratio
- RBE: dose ratio at isoeffect

(iv) Dose-response sketches

- Single hit: Linear ($\ln S = -\alpha D$), no threshold
- Multiple hit: Sigmoidal, shoulder at low dose
- Adaptive: Biphasic – initial hypersensitivity at low dose, plateau or induced resistance at higher dose

14.3.2 (b) Bragg Peak

(i) Spread-Out Bragg Peak (SOBP) and RBE



- Depth-dose plot: Physical dose is flat in SOBP, falls sharply distally.
- RBE: Low at entrance (1), rises along SOBP, peaks at distal end (higher LET).

Relation: Biological dose = physical dose $\times$ RBE. LET increases towards the distal end, so RBE rises, requiring careful treatment planning to achieve uniform effect.

(ii) Comparison of RBE: Proton vs C-12

- Proton: RBE ≈ 1.1 , nearly constant, slight rise at distal Bragg peak (1.5)
- C-12: RBE 1-2 at entrance, rises to 2-3+ in SOBP, peaks distally due to higher LET

(iii) **Dose-mean lineal energy (y_D)** Dose-mean lineal energy y_D quantifies energy deposition in microdosimetry:

$$y_D = \frac{\int_0^\infty y^2 f(y) dy}{\int_0^\infty y f(y) dy}$$

where $f(y)$ is the frequency distribution; $d(y) = y f(y) / \langle y \rangle$ is the dose distribution.

14.3.3 (c) Metastases and Radionuclides

(i) Development stages and suitable radionuclides

1. **Local invasion:** Primary tumor invades ECM; suitable: local alpha emitters (^{223}Ra , ^{211}At)
2. **Intravasation/circulation:** Cells in blood/lymph; suitable: systemic beta emitters (^{177}Lu -PSMA, ^{90}Y -microspheres)
3. **Extravasation/arrest:** Cells adhere at distant sites; suitable: uptake-specific radionuclides (^{131}I for thyroid)
4. **Colonization/proliferation:** Micrometastases grow; suitable: alpha for small mets (^{223}Ra , ^{225}Ac), beta for larger (^{177}Lu , ^{90}Y)

(ii) Advantages and disadvantages

- Advantages: Targeted, treats multifocal disease, theranostic, minimally invasive, outpatient possible
- Disadvantages: Off-target toxicity (kidney/bone marrow), variable uptake, radiation safety, high cost, limited to target-expressing cancers

14.3.4 (d) Boron Neutron Capture Therapy (BNCT)

(i) **Estimate absorbed dose at 2 cm depth** Given: Total fluence $\Phi_0 = 3 \times 10^{14}$ n/cm², attenuation factor 10 per cm, kerma factor $f_N = 2.74 \times 10^{-13}$ Gy cm²/n.

$$\Phi(z) = \Phi_0 \cdot 10^{-z} \quad (z \text{ in cm})$$

$$D(2 \text{ cm}) = \Phi(2) \cdot f_N = (3 \times 10^{14} \cdot 10^{-2}) \cdot 2.74 \times 10^{-13} \approx 0.82 \text{ Gy}$$

Surface dose $D_0 \approx 82.2 \text{ Gy}$.

(ii) **Physics explanation** Dose decreases with depth due to exponential attenuation of thermal neutrons: $\Phi(z) \propto e^{-\Sigma z}$, where $\Sigma \approx \ln 10 \approx 2.3 \text{ cm}^{-1}$. Thermal neutrons deposit energy via capture reactions (mainly $^{14}\text{N}(n,p)^{14}\text{C}$) producing short-range charged particles. Dose = local fluence $\times$ kerma. Reduced fluence at depth $\rightarrow$ reduced dose.

14.4 Q4

14.4.1 a) Cells and Radiation Effects

(i) **Three Types of Cell Change** Radiation can induce various cellular changes, leading to different modes of cell death or dysfunction:

- **Mitosis (Mitotic Death):** DNA damage leads to chromosomal aberrations that manifest during cell division. Cells attempt to divide but fail due to genomic instability, resulting in delayed cell death. Common in rapidly dividing cells.
- **Apoptosis (Programmed Cell Death):** Active, energy-dependent self-destruction in response to radiation. Characterized by caspase activation, DNA fragmentation, and cell shrinkage without inflammation. More prevalent in lymphocytes and at lower doses.
- **Necrosis:** Passive, uncontrolled cell death caused by severe radiation damage overwhelming repair mechanisms. Leads to cell swelling, membrane rupture, and release of intracellular contents, causing inflammation. Typically occurs at high doses or in hypoxic conditions.

(ii) Stochastic vs Deterministic Effects

- **Stochastic Effects:** Probabilistic; probability increases with dose, no threshold exists, severity independent of dose. Examples: cancer, genetic mutations. Manifest years later and evaluated using linear-no-threshold (LNT) models.
- **Deterministic Effects:** Threshold dose exists; severity increases with dose above threshold. Examples: skin erythema, cataracts, sterility. Manifest predictably, acutely or chronically, depending on organ and dose.

14.4.2 b) Relative Biological Effectiveness (RBE) and Quality Factor

(i) **Classical Definition of RBE** The *Relative Biological Effectiveness (RBE)* is defined as:

$$\text{RBE} = \frac{D_{\text{reference}}}{D_{\text{test}}} \quad \text{for the same biological effect.} \quad (18)$$

Using the linear-quadratic (LQ) model for cell survival:

$$S = \exp(-\alpha D - \beta D^2), \quad (19)$$

where α represents linear (irreparable) damage and β represents quadratic (repairable) damage.

For radiation A (low LET) and B (high LET), equating survival:

$$\alpha_A D_A + \beta_A D_A^2 = \alpha_B D_B + \beta_B D_B^2, \quad (20)$$

so that

$$\text{RBE} = \frac{D_A}{D_B}. \quad (21)$$

For small doses ($D \rightarrow 0$, βD^2 negligible):

$$\alpha_A D_A \approx \alpha_B D_B \Rightarrow \text{RBE} \approx \frac{\alpha_B}{\alpha_A}, \quad (22)$$

which is independent of dose.

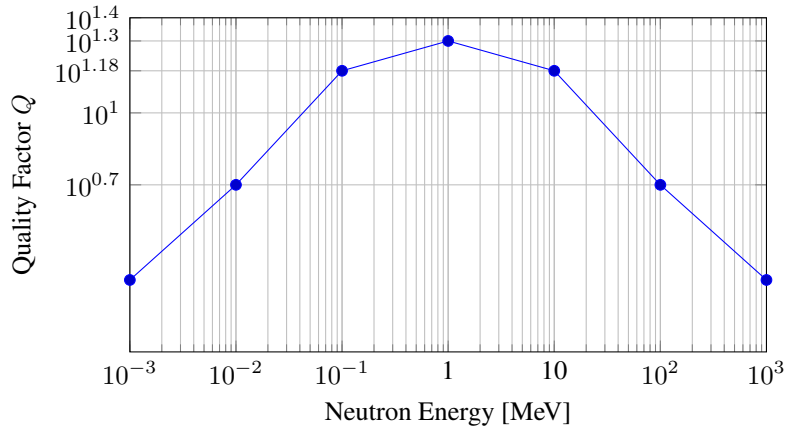


Figure 3: Neutron quality factor Q as a function of neutron energy.

(ii) Neutron Quality Factor Q vs Energy **Explanation:** Q is low (2–5) for thermal neutrons, peaks (20) at 0.5–1 MeV due to elastic scattering producing high LET recoil protons, and decreases at higher energies as forward-scattered particles produce lower ionization density.

14.4.3 c) Relative Depth Dose in Water Phantom

(i) 14 MeV Neutrons vs 6 MV Photons

- **6 MV X-rays:** Surface build-up to $d_{\max} \sim 1.5$ cm, then exponential fall-off. 50% dose at 15–20 cm.
- **14 MeV neutrons:** Higher skin dose, flatter initial curve due to scatter, 50% dose at 10–15 cm. Poorer penetration than photons.

(ii) Fast Neutrons (Be(d,48.5), Be(p,48.5)) vs 18 MV Photons

- **Be(d,48.5) and Be(p,48.5):** Neutron spectra up to 50 MeV, mean 20 MeV; 50% dose at 15–20 cm.
- **18 MV photons:** Deeper $d_{\max} \sim 3$ cm, 50% dose at 25 cm, lower skin dose than neutrons.

(iii) Proton vs Antiproton Beams

- **Protons:** Inverse depth dose with Bragg peak, low entrance dose, zero beyond.
- **Antiprotons:** Bragg peak plus annihilation products (pions, gammas), higher peak dose (2–3x), slight tail beyond peak, higher biological effectiveness.

(iv) Heavy Ion Therapy (HIT) vs Proton Therapy (PT)

- Both: Bragg peak at same range.
- HIT (e.g., ¹²C): Sharper peak, higher LET and RBE (2–3), minimal distal tail.
- Treatment planning systems (TPS): Pencil beam algorithms, Monte Carlo simulations (FLUKA, GEANT4), modeling nuclear fragmentation, multiple scattering, energy loss (Bethe-Bloch), SOBP optimization.

14.4.4 d) Protons, IMRT, and Heavy Ions

(i) Proton Beam Delivery Methods

- **Passive Scattering:** Uses scatterers and modulators; patient-specific collimators required.
- **Active Scanning (Pencil Beam):** Magnetic steering, layer-by-layer deposition, intensity modulation.
- **Uniform Scanning:** Hybrid approach; scans with uniform dose per layer.

Advantages: High conformality, reduced hardware neutrons, flexible dose painting. **Disadvantages:** Sensitive to organ motion (interplay effects), longer treatment times, requires robust optimization.

(ii) Out-of-Field Dose Contributions

- **IMRT (photons):** Head leakage, collimator scatter, patient scatter; neutrons generated at >10 MV.
- **Protons:** Secondary neutrons from beamline (passive > scanning) and nuclear interactions in patient.
- **Heavy Ions:** Neutrons and charged fragments from projectile/target; generally lower out-of-field dose than passive protons.

(iii) Neutron Sources in Proton Therapy

- Beamline components in passive systems (primary source).
- Nuclear interactions in patient (unavoidable).

Minimization: Use active scanning to reduce hardware-induced neutrons; patient-induced neutrons remain (10–20% of total). Fully avoiding neutrons is not possible.

14.5 Q5

14.5.1 a) Carcinogenesis, DNA repair mechanisms and Targeted Alpha Therapy (TAT)

Carcinogenesis is the multistep process by which normal cells transform into malignant cells through accumulation of genetic and epigenetic alterations. Causes include ionising radiation, chemical carcinogens, viruses, inherited mutations and chronic inflammation. **Examples:**

- Lung cancer from tobacco carcinogens.
- Skin cancer from ultraviolet (UV) radiation.
- Leukaemia following exposure to ionising radiation.

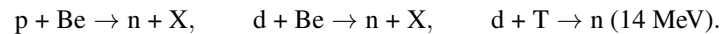
Common DNA repair pathways (brief):

- **Base excision repair (BER):** Corrects small base lesions (oxidation, alkylation). Key steps: glycosylase removes base → AP endonuclease cuts backbone → polymerase/ligase fill-and-seal.
- **Nucleotide excision repair (NER):** Removes bulky helix-distorting lesions (e.g. UV-induced pyrimidine dimers) by dual incision, excision of oligonucleotide, resynthesis and ligation.
- **Mismatch repair (MMR):** Corrects replication mispairs and insertion/deletion loops; improves replication fidelity.
- **Homologous recombination (HR):** Error-free repair of double-strand breaks (DSBs) using sister chromatid as template (S/G2 phase).
- **Non-homologous end joining (NHEJ):** Direct ligation of DSB ends (active in G1), relatively error-prone.
- **Single-strand break repair (SSBR):** Similar machinery to BER for single-strand breaks.
- **Principle:** Alpha-emitting radionuclides are conjugated to targeting vectors (antibodies, peptides, small molecules) that selectively bind tumour-associated antigens. Short-range (tens of μm) high-LET alpha particles deposit dense ionisation locally, producing complex DNA damage (clustered DSBs) and high cell kill, while sparing surrounding normal tissue.
- **Clinical rationale:** Ideal for micrometastases, single-cell disease or minimal residual disease because of very high cell-killing per decay and short range.
- **Common alpha emitters:**
 - ^{211}At ($t_{1/2} = 7.2 \text{ h}$)
 - ^{213}Bi ($t_{1/2} = 46 \text{ min}$)
 - ^{225}Ac ($t_{1/2} = 10 \text{ d}$) — multiple alpha daughters
 - ^{223}Ra ($t_{1/2} = 11.4 \text{ d}$) — used for bone metastases (RaCl_2)
 - Others: $^{212}\text{Pb}/^{212}\text{Bi}$, ^{227}Th , ^{149}Tb .
- **Challenges:** Radiochemistry (stable conjugation, daughter recoil), supply/production, dosimetry (microdosimetry), normal-organ toxicity from daughter redistribution.

14.5.2 b) Neutron sources for radiotherapy and spectra

14.5.3 (i) Fast neutron therapy (FNT)

Production reactions: Typical accelerator reactions producing therapeutic fast neutrons include



Proton vs deuteron projectiles:

- **Deuteron-induced (d + Be):** Tend to give higher neutron yield and a harder (higher-energy) forward-peaked spectrum than protons of same nominal beam energy, because the deuteron can break up and the neutron continues forward with substantial energy.
- **Proton-induced (p + Be):** Typically lower yield and somewhat softer spectrum for the same accelerator terminal voltage.

Measuring neutron and gamma absorbed doses in FNT (example method):

1. Use a pair of tissue-equivalent (TE) ionisation chambers: one TE-chamber (sensitive to both neutrons and photons) and one photon-sensitive chamber (e.g. graphite or air cavity with gamma calibration).
2. Measure total dose with TE chamber, measure gamma dose with photon-calibrated chamber; neutron dose estimated as difference after applying chamber response corrections and calibration factors:

$$D_n \approx D_{\text{TE,meas}} - D_{\gamma,\text{meas}} \times C_{\gamma \rightarrow \text{TE}}$$

where $C_{\gamma \rightarrow \text{TE}}$ corrects for TE chamber photon response.

3. Complementary detectors: activation foils (for spectral info), proton-recoil spectrometers, Bonner sphere system for spectrum unfolding.

Schematic depth-dose behaviour (qualitative):

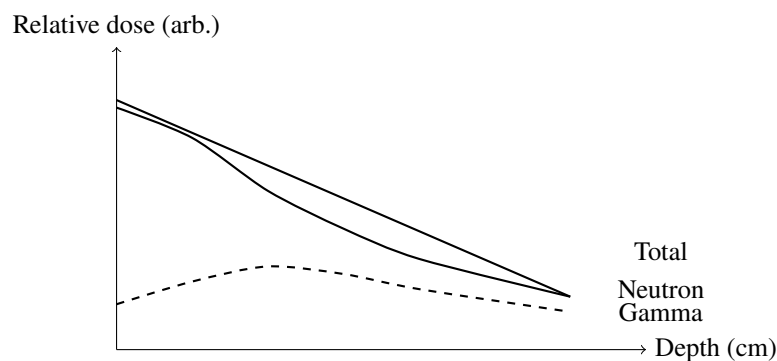


Figure 4: Qualitative sketch of neutron and gamma contributions versus depth in water for a fast-neutron beam.

Explanation: Fast neutrons produce high entrance dose (no pronounced build-up like MV photons). Neutron dose decreases with depth due to attenuation and energy loss; gammas (from neutron interactions and capture) contribute increasingly relative to neutrons at greater depths.

14.5.4 (ii) BNCT beam production

Methods:

- **Reactor-based:** Thermal or epithermal neutrons produced in research reactors, moderated and filtered to desired energy distribution (e.g. Bi, Al filters, moderators such as D₂O or graphite).
- **Accelerator-based:** Proton or deuteron beams on light targets (Li, Be, or B) produce neutrons via (p, n) or (d, n) reactions — increasingly used to bring BNCT into hospital settings.

Pros/cons:

- **Reactor:** very high flux (advantageous for shorter treatment times and deep penetration), but large infrastructure, regulatory and access issues.
- **Accelerator:** compact, can be sited in hospital, no fissile material, but lower neutron flux and target cooling/engineering challenges.

Advantage of epithermal neutron beams: Epithermal neutrons (approx. 0.5 eV–10 keV) penetrate several centimetres before thermalising; they thermalise within tissue, increasing thermal flux at depth and enabling effective $^{10}\text{B}(n, \alpha)^7\text{Li}$ capture within deep-seated tumours while reducing excessive skin dose compared with purely thermal beams.

14.5.5 (iii) Boron dose distribution: thermal vs epithermal BNCT

- **Thermal BNCT:** High thermal neutron flux at surface $\Rightarrow$ high skin dose, poor depth penetration; suitable only for superficial lesions.
- **Epithermal BNCT:** Epithermal neutrons enter and thermalise in tissue, producing a thermal flux peak at depth (a few cm), yielding higher boron-capture dose in tumours located several cm beneath the surface and lower entrance dose.

14.5.6 (iv) Components of radiation fields in BNCT

Components present in clinical BNCT beams:

- Thermal and epithermal neutrons (therapeutic component via $^{10}\text{B}(n,\alpha)^7\text{Li}$).
- Fast neutrons (contaminant; produce recoil protons and undesired dose).
- Gamma rays (from source, moderators and capture reactions; contribute to background dose).
- Prompt charged particles from capture reactions (the alpha and ^7Li from boron capture are the therapeutic local contributors).

14.5.7 (v) Absorbed and biological depth-dose in epithermal BNCT (schematic)

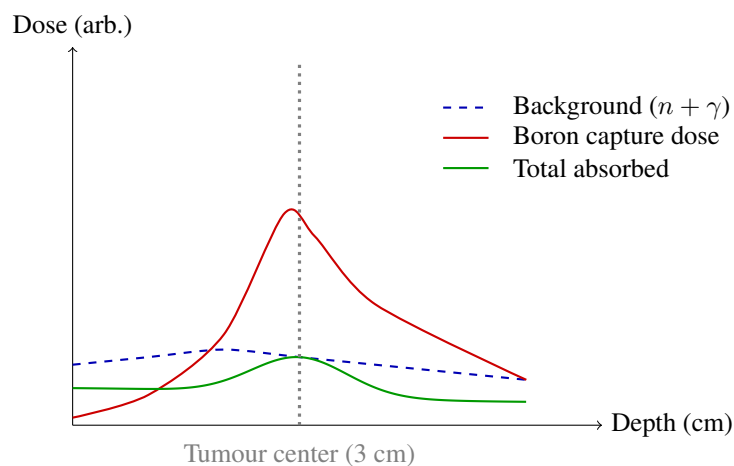


Figure 5: Schematic absorbed and boron-capture dose versus depth for epithermal BNCT with a tumour at 3 cm. Biological (RBE-weighted) dose accentuates the boron peak because alpha/ ^7Li products have high RBE.

14.6 Q6

14.6.1 a) Major sources of background radiation

- **Cosmic rays:** Extraterrestrial origin; dose depends on altitude and geomagnetic latitude.
- **Terrestrial radioactivity:** Radon (from U/Th decay chains), K-40, U and Th series in soil and building materials.
- **Internal:** Naturally occurring radionuclides in the body (e.g. K-40, C-14).
- **Man-made:** Medical exposures (diagnostic radiology, CT), occupational and small contributions from nuclear industry and fallout.

14.6.2 b) Deterministic vs stochastic effects (recap)

- **Deterministic:** Threshold exists; severity increases with dose (e.g. skin erythema, cataract, radiation burns, acute radiation syndrome).
- **Stochastic:** No threshold (practical model); probability of occurrence increases with dose while severity is independent of dose (e.g. radiation-induced cancer, heritable genetic effects).

14.6.3 c) Dose equivalent and effective dose

- **Dose equivalent:** $H_T = \sum_R w_R D_{T,R}$, where $D_{T,R}$ is absorbed dose in tissue T from radiation R and w_R is radiation weighting factor (quality factor proxy). Unit: sievert (Sv).
- **Effective dose:** $E = \sum_T w_T H_T$, where w_T are tissue weighting factors representing relative radiosensitivity. Used to estimate stochastic risk for non-uniform exposures and to set protection limits.

14.6.4 d) (i) Methods for separate gamma and neutron dosimetry in mixed fields

Common approaches:

- **Paired detectors:** TE ionisation chamber (sensitive to both) and photon-only chamber (e.g. graphite); subtract corrected photon contribution to obtain neutron dose.
- **Neutron counters:** ^3He or BF_3 proportional counters (thermal-neutron sensitive) often used with moderators and energy-dependent response correction.
- **Spectrometers/Bonner spheres:** Unfold spectrum and compute dose components.
- **Electronic discrimination:** Some detectors (scintillators with pulse-shape discrimination) separate neutron and gamma events.

14.6.5 d) (ii) Activation foils and albedo dosimetry

Activation foils: Expose multiple foils (Au, In, Mn, Al, etc.) with well-known cross-sections; measure induced activity and solve inverse problem (spectrum unfolding) to derive neutron flux $\phi(E)$ and dose. Advantages: energy sensitivity, integrated measurement; disadvantages: offline, time-consuming, requires decay-corrections and unfolding algorithms.

Albedo dosimetry: Uses TLDs or other dosimeters to measure neutrons scattered (albedo) back from the body or moderator. Advantage: personal, simple; disadvantage: strongly energy dependent, calibration complex, less accurate for high-energy neutrons.

14.6.6 e) (i) Main contributions to neutron tissue KERMA in different energy ranges

KERMA (kinetic energy released per unit mass) into charged particles in tissue arises mainly from:

- **Thermal and epithermal region ($\ll \text{keV}$ – keV):** Capture reactions (e.g. $\text{H}(n,\gamma)$) and small recoils — low KERMA per incident neutron.
- **Resonance and fast region (keV – MeV):** Elastic scattering on hydrogen produces recoil protons — major contributor to KERMA at many therapeutic energies.
- **High-energy region ($>20 \text{ MeV}$):** Inelastic reactions, (n,p) , (n,α) and spallation produce secondary charged particles, increasing KERMA and modifying spectral composition.

14.6.7 e) (ii) Energy dependence of neutron KERMA (qualitative sketch)

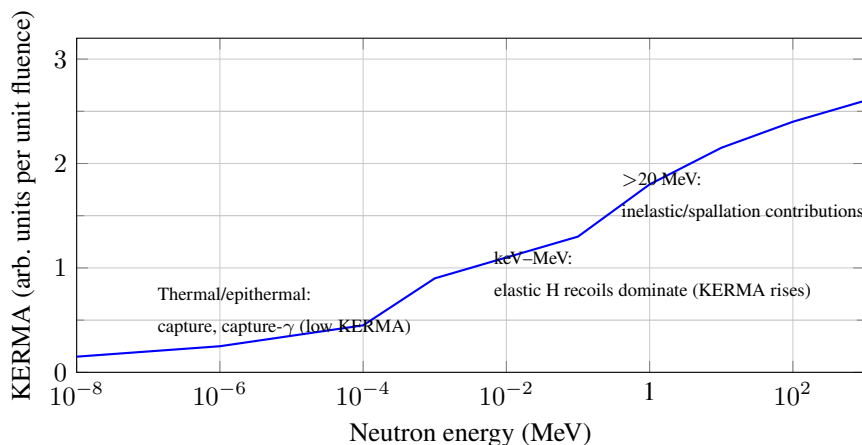


Figure 6: Qualitative KERMA (per unit fluence) vs neutron energy for tissue. Energy regions are labelled with dominant interaction types.

14.6.8 e) (iii) Real-time noninvasive 3D dosimetry methods (BNCT, proton, heavy ion)

- **Prompt-gamma imaging:** Detect prompt gamma rays from nuclear interactions (e.g. proton-induced prompt gammas) with Compton cameras or collimated detectors to infer beam range and deposited dose in near-real-time.
- **PET imaging of activated species:** For proton/ion therapy, positron-emitting isotopes produced in tissue (e.g. ^{15}O , ^{11}C) are imaged after or during irradiation for range verification (not strictly real-time).
- **In-beam prompt gamma / prompt neutron detection for BNCT:** Measurement of 478 keV gamma from $^{10}\text{B}(n,\alpha)^7\text{Li}$ (when present) and other prompt signals to infer boron-capture distribution.

These techniques provide non-invasive verification of range and dose distribution, but require sophisticated detectors, corrections for background and complex reconstruction algorithms.

I. Boron Neutron Capture Therapy (BNCT)

1) BNCT Beam Production Methods

(a) Reactor-based neutron sources

- **Thermal neutron beam** – Reactor moderator directly provides large thermal flux. **Advantage:** high intensity, stable spectrum. **Disadvantage:** shallow penetration, unsuitable for deep tumours.
- **Epithermal neutron beam** – Uses filtering/moderator assembly to shift spectrum to 1 eV–10 keV. **Advantage:** optimal penetration and depth-dose profile for deep-seated tumours. **Disadvantage:** Complex filtering, gamma contamination.

(b) Accelerator-based BNCT (AB-BNCT)

- Proton beam hits target (e.g., Li, Be) → neutron-producing nuclear reactions.
- **Advantages:** Can be hospital-based, shutdown possible, more compact.
- **Disadvantages:** Lower flux, engineering complexity, heat management.

Advantage of Epithermal Neutrons

Epithermal neutrons (1 eV–10 keV) can penetrate a few centimeters into tissue before slowing down to thermal energies, allowing:
maximum thermal neutron concentration at the tumour depth.

Thus they enable treatment of deep tumours (e.g., 3–7 cm), whereas thermal neutrons only treat superficial lesions.

Method for Epithermal Neutron Production in Reactors

- Begin with thermal neutron field.
- Use a **beam shaping assembly (BSA)**:
 - Moderators (Al, MgF_2 , Teflon)
 - Filters (S, Ti)
 - Reflectors (Be, graphite)
 - Collimators

to shift the spectrum upward to the epithermal range.

Comparison of Boron Dose in Thermal vs Epithermal BNCT

Thermal BNCT:

- Boron dose peaks at skin and drops rapidly.
- Suitable only for shallow tumours.

Epithermal BNCT:

- Epithermal neutrons travel deeper then thermalize at tumour depth.
- Boron dose peaks at depth (e.g., 3 cm).

Radiation Field Components in BNCT

BNCT delivered dose consists of:

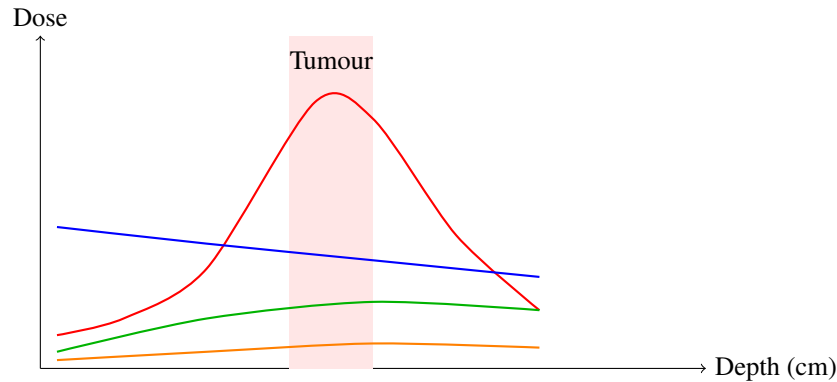
D_B : Boron dose from $^{10}\text{B}(n, \alpha)^7\text{Li}$,

D_H : Hydrogen dose from n -proton elastic scattering,

D_N : Nitrogen dose from $^{14}\text{N}(n, p)$,

D_γ : Gamma-ray dose.

Depth-Dose Sketch for a 3 cm Deep Tumour



II. Radiobiology

1) Target Theory

Target theory states that radiation kills a cell if a sensitive vital structure (the “target”) is inactivated by radiation.

Most likely target: cell DNA.

2) Acute vs Late Effects

Acute effects

- Occur within days–weeks
- In rapidly dividing tissues
- Example: **erythema**

Late effects

- Occur months–years later
- In slowly dividing tissues
- Example: **radiation-induced cancer**

3) Linear Quadratic (LQ) Model

Cell survival:

$$S = e^{-(\alpha D + \beta D^2)}$$

α : single-track events, β : two-track events

Alpha/Beta ratio:

$$\frac{\alpha}{\beta}$$

Dose at which linear and quadratic components are equal; determines fractionation sensitivity.

4) Therapeutic Ratio

$$\text{Therapeutic Ratio} = \frac{\text{Normal tissue tolerance}}{\text{Tumour control dose}}.$$

5) Stochastic Effect

Correct answer:

(b) has a higher probability of occurring with an increase of dose

No threshold; severity independent of dose.

6) Best Example of a Late Effect

Correct answer:

(g) Cancer

III. Radiation Protection

1) Radiation Protection Principles

(i) Principles

- **Justification:** benefit must outweigh risk.
- **Optimisation (ALARA):** exposures kept as low as reasonably achievable.
- **Dose Limitation:** for non-medical exposures only.

(ii) International Group and Document Issued by:

ICRP – International Commission on Radiological Protection

ICRP Publication 60 (1991) and 103 (2007)

(iii) Why no dose limits for medical exposure? Because:

- Limiting dose may compromise diagnosis or treatment.
- Medical exposures are **justified**, not subject to limits.

2) IAEA BSS (2014)

(i) Aims

- Provide global radiation safety requirements.
- Protect workers, public, and patients.
- Ensure consistent application across nations.

(ii) Relation between ICRP, IAEA BSS, NSW EPA

- **ICRP:** Provides scientific recommendations.
- **IAEA BSS:** Converts ICRP advice into regulatory requirements.
- **NSW EPA:** Implements regulations in New South Wales based on BSS.

3) Dose Limits and Second Cancer

(i) ICRP 60 Dose Limits for Radiation Workers

Annual limit: 20 mSv/year (averaged over 5 years)

Five-year limit: 100 mSv

(ii) ALARA Principle

- Exposures kept as low as reasonably achievable considering:

1. **Economic factors**
2. **Social factors**

(iii) Criteria for Radiation-Induced Second Cancer A tumour is classified as radiation-induced if:

1. Located within irradiated field

2. Sufficient latency period (e.g., 5–10 years)
3. Different histology from the primary tumour
4. Dose must be high enough to induce malignancy
5. Patient must have been disease-free in between